

An Optimal Quantum Query Lower Bound for Matrix-Product Verification

Shiva Kintali*

September 20, 2026

Abstract

We prove that deciding whether $AB = C$ for three $n \times n$ matrices has bounded-error quantum query complexity $\Omega(n^{5/3})$ in the exact whole-entry oracle model, over every fixed nonzero associative ring with identity. Our lower bound matches the Buhrman–Špalek quantum-walk upper bound, establishing its optimality in the exact whole-entry query model.

1 Introduction

Matrix-product verification asks whether $AB = C$ without computing the product. Freivalds’s randomized test uses $O(n^2)$ field operations [11]. We prove that the $O(n^{5/3})$ quantum query bound is optimal over every fixed nonzero associative ring with identity, combining a common source-hardness mechanism with distinct finite-field and integer masking constructions.

Our main idea is residual transfer: after averaging a bounded function of coordinate degree L over factors with fixed residual $AB - C$, we approximate its residual dependence by a sum of terms, each supported on $O(L/n)$ rows and columns. Equality-matrix substitution then transfers element-distinctness hardness. Over finite fields the constants are uniform in the field size; over the integers the hard instances have $O(n)$ -bit entries. The bound also holds for the negative-weight adversary and under the promise that at most one entry of C is wrong. Over finite fields and finite integer boxes, it also holds on average under a balanced distribution on exact products and products with one uniformly located unit error.

Related work. Buhrman and Špalek [8] improved an unpublished $O(n^{7/4})$ algorithm of Ambainis, Buhrman, Høyer, Karpinski, and Kurur to $O(n^{5/3})$ queries. Their stored-submatrix query algorithm works over rings. Their original time-efficient analysis is stated for integral domains [8]; extensions to rings are discussed in [15]. They proved an $\Omega(n^{3/2})$ lower bound over $\mathbb{F}_2$ by embedding an OR of parities; this exponent also extends to nonzero unital rings [15]. They asked whether it could be improved and whether a single wrong entry is harder to detect [8, Section 6.3]. Their algorithm builds on quantum walks [4, 22, 17].

Positive-weight adversaries encounter a certificate barrier: every such bound for a total function is at most $\sqrt{C_0(f)C_1(f)}$ [21, Theorem 3.2]. Here a NO input has a certificate consisting of one row of A , one column of B , and the corresponding entry of C , whereas all $3n^2$ entries form a YES certificate. Thus $C_0(f) \leq 2n + 1$ and $C_1(f) \leq 3n^2$, giving an $O(n^{3/2})$ barrier. Our proof uses real coordinate-degree lower bounds [6, 1, 3] and negative-weight adversaries [12, 16]. Bun and Thaler

*Email: shiva.kintali@gmail.com Homepage: shivakintali.github.io

[9] construct explicit dual polynomials for collision and element distinctness; Belovs [7] converts dual witnesses directly into adversaries. We use a finite linear program for the exact dual correction and give its assignment-isometry specialization explicitly.

The integer argument also uses geometry of numbers: Ball’s cube-slicing theorem [5, 18], the kernel-covolume identity, and Minkowski’s second theorem [2, Section 2]. Rectangular-unimodular density results [19] suggest the surjectivity step, but a finite prime sieve and quantitative short-lift estimate are needed here to obtain an explicit $O(n)$ -bit hard restriction. Boolean OR–AND matrix multiplication is a different problem [10, 15]; none of our ring reductions identifies it with ordinary multiplication.

1.1 Exact entry access

Throughout, $[n] = \{1, \dots, n\}$, all logarithms are natural unless marked $\log_2$, and products use the indicated order $(AB)_{ij} = \sum_k A_{ik}B_{kj}$. A ring is fixed, associative, and has identity $1_R \neq 0$, unless a different hypothesis is stated. Let $\text{MPV}_R(A, B, C) = \mathbf{1}[AB = C]$. For the addressed list z of the $3n^2$ entries, the exact-value oracle is

$$O_z|t, b, w\rangle = |t, b + z_t, w\rangle, \quad t \in [3n^2], \quad b \in R. \quad (1)$$

The response register is $\ell_2(R)$, with basis indexed by ring elements. Translation is unitary, and conjugating by $b \mapsto -b$ implements the inverse oracle. A controlled query can be simulated with two queries: read the addressed value into a clean response ancilla, add it to the original response register conditional on the control, and erase the ancilla with an inverse query. Consequently, three separate entry oracles and this addressed oracle are equivalent up to a constant factor.

Algorithms are finite query circuits with input-independent unitaries and a final measurement. Finite-outcome intermediate measurements and classically controlled stopping are also allowed, with a finite history tree and a worst-case query cap. Exact arithmetic and equality tests between queries have no query cost. The worst-case query count of such an algorithm is the maximum number of queries along any history that occurs with positive probability on some required input. The notation $Q_{1/3}$ denotes the least worst-case query count of an algorithm with success probability at least $2/3$ on every required input. For a promise algorithm with worst-case cap T , halt with an arbitrary output immediately before any attempted $(T + 1)$ st query. This retains all intervening gates and measurements and makes acceptance defined and bounded on all oracle inputs without changing promised behavior. On a finite restriction, the usual XOR oracle for fixed-length entry labels simulates (1) with two queries: write the label into a clean ancilla, translate the original response register by the decoded value, and erase the label. The same construction implements an inverse or controlled query and allows an arbitrary original response register. For infinite rings this is an ideal exact-value model: $\ell_2(R)$ may be nonseparable, and no finite-qubit, finite-time, or finite-precision implementation is asserted. The algorithms never require a uniform superposition over R .

All lower bounds have finite support. For any finite product domain $\Omega = \prod_t \Sigma_t$, let $V_L(\Omega)$ be the real span of functions depending on at most L entry coordinates. This is equivalently degree at most L in the one-hot indicators $\mathbf{1}[z_t = a]$. The bounded approximate coordinate degree $\deg_{\text{bd}, 1/3}(f)$ is the least such degree of a real function p satisfying $0 \leq p \leq 1$ on the entire product domain and $|p - f| \leq 1/3$ on every required input. Function sup norms are taken over that entire domain, including off-promise inputs. This degree is not algebraic degree over the underlying ring. The finite negative-weight adversary value is defined in Section 6.4; for infinite R , $\text{Adv}^\pm(\text{MPV}_R)$ denotes its supremum over finite restrictions.

1.2 Main results

Theorem 1.1 (Optimal query complexity over unital rings). *For every fixed nonzero associative ring R with identity, in the exact whole-entry model,*

$$\text{Adv}^\pm(\text{MPV}_R) = \Omega(n^{5/3}), \quad Q_{1/3}(\text{MPV}_R) = \Theta(n^{5/3}).$$

The lower bound is witnessed on a finite restriction. The implicit constants are absolute: they depend neither on n nor on R .

The query-complexity assertion is proved in Section 5, completing the main lower-bound argument. The adversary assertion is proved separately in Section 6.6, after the stronger finite-domain bounds have been established.

The two arithmetic ingredients have stronger quantitative statements. For a finite field put $\Omega_{n,q} = (\mathbb{F}_q^{n \times n})^3$. In both settings write $f(A, B, C) = \mathbf{1}[AB = C]$.

Theorem 1.2 (Finite fields, uniformly in field size). *There are absolute $c > 0$ and n_0 such that for every finite field $\mathbb{F}_q$ and $n \geq n_0$,*

$$\widetilde{\text{deg}}_{\text{bd},1/3}(f|_{\Omega_{n,q}}) \geq cn^{5/3}, \quad Q_{1/3}(f|_{\Omega_{n,q}}) = \Theta(n^{5/3}).$$

All constants are independent of q , which may grow with n .

Theorem 1.3 (Integers with linear entry bit length). *There are absolute $c, b > 0$ and n_0 such that for every $n \geq n_0$ there is a finite product restriction $\mathcal{X}_n$ of integer matrix triples, with at most bn bits per entry, for which*

$$\widetilde{\text{deg}}_{\text{bd},1/3}(f|_{\mathcal{X}_n}) \geq cn^{5/3}, \quad Q_{1/3}(f|_{\mathcal{X}_n}) = \Theta(n^{5/3}).$$

Thus the lower bound already holds on hard instances with $O(n)$ -bit entries. The explicit boxes and a uniform extension to larger boxes are given in Section 4.

Let E_{ij} have entry 1 at (i, j) and zero elsewhere, interpreted in the stated arithmetic domain.

Theorem 1.4 (Adversary, singleton, and distributional bounds). *In both settings of Theorems 1.2 and 1.3, the negative-weight adversary bound is $\Omega(n^{5/3})$, also under the promise*

$$AB - C \in \{0\} \cup \{E_{ij} : i, j \in [n]\}.$$

The same query and adversary lower bounds hold for the larger promise that at most one entry is wrong.

For the finite-field case, choose A, B independently and uniformly. For the integer case, use the integer width H specified in Section 4 and choose all factor entries independently and uniformly from $[-H, H] \cap \mathbb{Z}$. With probability $1/2$ set $C = AB$; otherwise choose (i, j) uniformly and set $C = AB - E_{ij}$. Average success at least $2/3$ under this balanced distribution requires $\Omega(n^{5/3})$ queries for algorithms with a worst-case query cap. Constants are uniform in q . All integer assertions also hold uniformly over the larger-box family defined in Section 4.

In particular, Theorem 1.1 applies to every field and to $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$, with exact entry access as defined above. The bit-length claim concerns the hard restriction, not a bit-time upper bound. For singleton residuals in composite characteristic, see Section 6.6.

Our proof strategy. The common mechanism averages $p(A, B, AB - D)$ while retaining the residual D . It maps bounded coordinate degree L to dependence on at most $K = O(L/n)$ rows and columns per residual Fourier term, with exponentially small uniform error. Substituting $D_{ij} = \mathbf{1}[x_i = y_j]$ gives source degree at most $2K$. A balanced split of an element-distinctness instance forces $K = \Omega(n^{2/3})$, hence $L = \Omega(n^{5/3})$. This proves the general-input lower bound without an adversary construction.

The two arithmetic constructions of residual transfer use different tools. Over finite fields, character orthogonality and random systematic-code distance give exact one-sided support restrictions. Over the integers, a prime sieve and short lattice lifts make low-coordinate observations nearly invariant under residual shifts. A signed correction applied to each row before tensoring turns this into an exact support restriction without paying a whole-input decomposition norm. In both cases, two separate factor averages are combined by a controlled row projection; the factors are never jointly conditioned. Entry-local encodings then transfer the lower bounds to every nonzero unital ring. Together with the stored-submatrix quantum walk, this completes the universal-ring query bound before the stronger conclusions are considered.

For the stronger conclusions, a collision sample produces a uniform zero-versus-singleton gap test. It controls one explicit signed seed; a finite LP corrects its low-degree moments, and the assignment isometries of [7] specify a negative-weight adversary. The construction is finite but not claimed to be computationally efficient. These averaging steps act on bounded functions and are not query-preserving algorithmic reductions. The same ring encodings subsequently transfer the finite adversary witnesses.

The transfer framework also yields $\Theta(ks^{2/3})$ queries for s -by- k times k -by- s verification when $k \geq s$, matching the rectangular upper bound of [8, Section 6.1].

Organization. Section 2 gives the common framework; Sections 3 and 4 supply its arithmetic ingredients. Section 5 completes the universal-ring query bound. Section 6 then proves the stronger distributional, singleton, and adversary bounds, and Section 7 treats rectangular verification and localization.

2 The common residual-transfer framework

The framework in this section has no field-specific hypothesis. Its input is a uniform approximation of residual averaging by functions with restricted row and column support. Sections 3 and 4 verify that input by different arguments.

2.1 Finite alphabets and polynomial degree

Fix a finite product domain $\Omega = \prod_{t=1}^{3n^2} \Sigma_t$ and a predicate $f : \Omega \rightarrow \{0, 1\}$. A partial assignment α fixes values of some input coordinates; $|\alpha|$ is their number and $z \sim \alpha$ means agreement. The empty assignment is allowed. Abbreviate $V_L(\Omega)$ to V_L when the domain is specified.

Lemma 2.1 (Coordinate-degree filtration). *The space V_L is the real span of indicators $\mathbf{1}[z \sim \alpha]$ with $|\alpha| \leq L$, and is exactly the space of one-hot polynomial functions of degree at most L on valid alphabet inputs. Fixing coordinates cannot increase this degree.*

Proof. Any function depending on coordinates U has the assignment expansion

$$F(z) = \sum_{a \in \prod_{t \in U} \Sigma_t} F(a) \prod_{t \in U} \mathbf{1}[z_t = a_t].$$

Conversely, products of one-hot indicators vanish when inconsistent and otherwise reduce to an assignment indicator. Fixing coordinates turns their indicator factors into constants. $\square$

Lemma 2.2 (Acceptance degree on finite restrictions). *A T -query algorithm in the model of Section 1.1 has a globally $[0, 1]$ -valued acceptance function in V_{2T} on every finite product restriction, also with infinite-dimensional workspace.*

Proof. On the restriction, the oracle and its inverse are finite sums

$$O_z^{\pm 1} = \sum_t \sum_{a \in \Sigma_t} \mathbf{1}[z_t = a] |t\rangle\langle t| \otimes \tau_{\pm a},$$

with identities on the remaining registers, where τ_a is translation by a . Induction on queries expresses the final state as a finite sum of indicator monomials of degree at most T times fixed Hilbert-space vectors. For the fixed acceptance effect M , the probability $\langle \psi_z, M\psi_z \rangle$ is a real polynomial of degree at most $2T$. Its range is $[0, 1]$ on all valid oracles. This is the finite-alphabet polynomial method [6], [3, Lemma 2.2]; no scalar coordinate basis for the workspace or infinite-domain averaging is needed.

For a finite adaptive history tree, apply the same induction to the unnormalized state along each history, including its fixed measurement operators. By retaining ancillary registers, initial states and measurement instruments may first be purified, so that each branch is represented by one fixed linear operator. A history with $t \leq T$ queries has state degree at most t and acceptance contribution of degree at most $2t$. Summing the finitely many accepting histories gives the same bound $2T$. No translation-invariant dummy response state or physical implementation of a controlled oracle is assumed. $\square$

We use the collision and element-distinctness degree bounds of Aaronson and Shi [1] in the polynomial formulation of [3, Definition 2.1 and Theorem 2.3]. Fix $0 < \varepsilon < 1/2$ and let N be even. If $M \geq 3N/2$, every real polynomial in the one-hot indicators $\mathbf{1}[u(i) = j]$ that lies in $[0, 1]$ on all maps $u : [N] \rightarrow [M]$, is at least $1 - \varepsilon$ on injective maps and at most ε on exactly two-to-one maps has degree $\Omega(N^{1/3})$. Every polynomial with the same global range bound that is at least $1 - \varepsilon$ on injective maps and at most ε on all non-injective maps has degree $\Omega(N^{2/3})$ when $M \geq N$; [3, Theorem 2.3] states this for $M = \Omega(N^2)$, and [3, Theorem 3.1] transfers the degree to every $M \geq N$. We use the element-distinctness bound with $M = 4N^2$. The implied constants depend only on ε . Crucially, boundedness is required on all legal maps, including off-promise maps, but not on invalid one-hot encodings.

2.2 Residual support and the transfer hypothesis

Let $\mathcal{D}_n = \{0, 1\}^{n \times n}$ and define $\mathcal{W}_K$ as the real span of functions depending only on $D[I, J]$, where $I, J \subseteq [n]$ and $|I|, |J| \leq K$. Equivalently, $\mathcal{W}_K$ is spanned by the Boolean characters

$$\chi_S(D) = (-1)^{\sum_{(i,j) \in S} D_{ij}}$$

whose supports meet at most K rows and at most K columns. Indeed every function on a rectangle has a Boolean Fourier expansion supported in that rectangle, and every such character depends on its row-by-column rectangle. In particular, constants belong to $\mathcal{W}_K$. This support description, rather than a choice of field character, is the common interface.

For each $D \in \mathcal{D}_n$, let μ_D be a probability measure on Ω supported on inputs satisfying $f(z) = \mathbf{1}[D = 0]$, and put

$$(\mathcal{R}p)(D) = \mathbb{E}_{z \sim \mu_D} p(z).$$

Thus $\mathcal{R}$ preserves constants and maps $[0, 1]$ -valued functions to $[0, 1]$ -valued functions. In the applications, μ_D is the law of $(A, B, AB - D)$ for the stated factor distribution. A Boolean residual is interpreted through the distinct ring elements 0, 1; integer multiplication is unchanged.

Definition 2.3 (Transfer hypothesis). For every fixed $a > 0$ and all sufficiently large n , suppose that, simultaneously for all integers $0 \leq L \leq an^{5/3}$, there is a real linear map

$$\mathcal{H}_L : V_L \longrightarrow \mathcal{W}_{\lfloor 16L/n \rfloor}$$

such that

$$\|\mathcal{R}p - \mathcal{H}_L p\|_\infty \leq \xi_{n,L} \|p\|_\infty, \quad 0 \leq \xi_{n,L} \leq e^{-cn} \quad (2)$$

for a constant $c > 0$ independent of a , L , and p ; the threshold on n may depend on a . The domain norm is on all of Ω and the output norm on all of $\mathcal{D}_n$. When the construction has auxiliary parameters, the constants and thresholds must be uniform in those parameters for uniform conclusions.

Corollary 2.4 (Normalization). Suppose Definition 2.3 holds. For $p \in V_L$ with $0 \leq p \leq 1$, write $g = \mathcal{R}p$, $h_0 = \mathcal{H}_L p$, $\xi = \xi_{n,L}$, $K = \lfloor 16L/n \rfloor$, and set

$$h = \frac{h_0 + \xi}{1 + 2\xi}. \quad (3)$$

Then $h \in \mathcal{W}_K$, $0 \leq h \leq 1$, and $\|h - g\|_\infty \leq 2\xi/(1 + 2\xi)$. For any probability measure ν on residuals,

$$h(0) - \mathbb{E}_\nu h(D) \geq \frac{g(0) - \mathbb{E}_\nu g(D) - 2\xi}{1 + 2\xi}. \quad (4)$$

Proof. All pointwise inequalities here and below are on the indicated finite domain. Since $g \in [0, 1]$ and $\|h_0 - g\|_\infty \leq \xi$, the range of h_0 is contained in $[-\xi, 1 + \xi]$. The constant function belongs to $\mathcal{W}_K$, so the normalization preserves the support restriction. Moreover,

$$h - g = \frac{h_0 - g + \xi(1 - 2g)}{1 + 2\xi},$$

which proves the uniform estimate. In a difference of two expectations the added constant cancels, and the two errors are each at most ξ ; this proves (4). $\square$

2.3 Equality substitution and element distinctness

For source lists $x, y \in [M]^n$, put

$$D(x, y)_{ij} = \mathbf{1}[x_i = y_j] \in \{0, 1\}.$$

Source labels do not need to embed in the target ring.

Lemma 2.5 (Degree of the pullback). For $h \in \mathcal{W}_K$, the function $h(D(x, y))$ has one-hot source-coordinate degree at most $2K$. If $0 \leq h \leq 1$ on $\mathcal{D}_n$, the same bounds hold for all list pairs, including lists with internal repetitions.

Proof. Each function of $D[I, J]$ becomes a function only of the labels x_i , $i \in I$, and y_j , $j \in J$. Their number is at most $2K$, so Lemma 2.1 gives the degree bound term by term. Every list pair produces a Boolean residual, which proves boundedness. $\square$

Lemma 2.6 (Source hardness). *If $h \in \mathcal{W}_K$, $0 \leq h \leq 1$, $h(0) \geq 0.98$, and $h(D) \leq 0.02$ for every nonzero $D \in \mathcal{D}_n$, then $K \geq c_1 n^{2/3}$ for an absolute constant $c_1 > 0$ and every $n \geq 1$.*

Proof. By Lemma 2.5, on source lists $x, y \in [M]^n$, $h(D(x, y))$ is a degree-at-most- $2K$ approximant to the predicate that the lists share no value. Take $M = 16n^2 = 4(2n)^2$. For $z \in [M]^{2n}$ and a permutation σ of $[2n]$, put $x^\sigma = (z_{\sigma(1)}, \dots, z_{\sigma(n)})$ and $y^\sigma = (z_{\sigma(n+1)}, \dots, z_{\sigma(2n)})$, and define

$$F(z) = \mathbb{E}_\sigma h(D(x^\sigma, y^\sigma)),$$

the average over a uniformly random permutation σ . Each indicator $\mathbf{1}[x_i^\sigma = a]$ is the single one-hot indicator $\mathbf{1}[z_{\sigma(i)} = a]$ of z , and the positions $\sigma(1), \dots, \sigma(2n)$ are distinct, so for each fixed σ the summand has degree at most $2K$ in the one-hot indicators of z . The law of σ does not depend on z , so F has degree at most $2K$ and takes values in $[0, 1]$. All-distinct inputs give $F \geq 0.98$. If the input has a repeated value, fix one equal pair of distinct positions. The two positions land in different lists with probability

$$2 \frac{n}{2n} \frac{n}{2n-1} = \frac{n}{2n-1} > 1/2.$$

On that event a cross collision exists, so the value is at most 0.02; on the complement it is at most one. Therefore $F \leq 1 - 0.98n/(2n-1) < 0.51$ on every colliding input. Its square satisfies

$$F^2 \geq 0.98^2 > 2/3 \quad \text{on distinct inputs,} \quad F^2 \leq 0.51^2 < 1/3 \quad \text{on colliding inputs,}$$

and has degree at most $4K$. Since $0 \leq F \leq 1$ on every legal source map, the same is true of F^2 . Thus F^2 approximates element distinctness on $[M]^{2n}$ with error below $1/3$ in the sense of [3, Definition 2.1], is bounded on all inputs, and $M = 4(2n)^2$. The element-distinctness degree bound recalled above, applied with source length $N = 2n$, gives $4K = \Omega((2n)^{2/3}) = \Omega(n^{2/3})$ for all sufficiently large n . For the remaining bounded range of n , note that $\mathcal{W}_0$ consists of the constant functions, so the separation of $h(0)$ from $h(D)$ forces $K \geq 1$; decreasing c_1 covers these n . $\square$

Proposition 2.7 (Transfer to the general-input lower bound). *Under Definition 2.3,*

$$\widetilde{\deg}_{\text{bd}, 1/3}(f) = \Omega(n^{5/3}), \quad Q_{1/3}(f) = \Omega(n^{5/3}).$$

The constants are uniform whenever the transfer constants are uniform.

Proof. Let $d = \widetilde{\deg}_{\text{bd}, 1/3}(f)$ and let $p \in V_d$ attain it, so that $0 \leq p \leq 1$ on Ω and $|p - f| \leq 1/3$ on every required input. Fix once and for all an odd integer s_0 with $e^{-s_0/18} \leq 0.01$, and consider the majority polynomial

$$\text{Maj}_{s_0}(t) = \sum_{j=(s_0+1)/2}^{s_0} \binom{s_0}{j} t^j (1-t)^{s_0-j} = \Pr[\text{Bin}(s_0, t) \geq (s_0+1)/2],$$

which lies in $[0, 1]$ for $t \in [0, 1]$. For $0 \leq t \leq 1/3$, Hoeffding's inequality [13] gives $\text{Maj}_{s_0}(t) \leq \Pr[\text{Bin}(s_0, t) - s_0 t \geq s_0/6] \leq e^{-s_0/18} \leq 0.01$, and symmetrically $\text{Maj}_{s_0}(t) \geq 0.99$ for $2/3 \leq t \leq 1$. Thus $p' = \text{Maj}_{s_0}(p) \in V_{s_0 d}$ is globally in $[0, 1]$ and approximates f with error at most 0.01 on required inputs. If $s_0 d > n^{5/3}$ the degree conclusion already holds. Otherwise use the transfer with $L = s_0 d$ and $a = 1$. For large n , its error is at most $1/200$. The support condition on μ_D gives $(\mathcal{R}p')(0) \geq 0.99$ and $(\mathcal{R}p')(D) \leq 0.01$ for $D \neq 0$. Corollary 2.4 therefore supplies h satisfying Lemma 2.6, with $K = \lfloor 16s_0 d/n \rfloor$. Consequently $16s_0 d/n \geq K \geq c_1 n^{2/3}$, so $d = \Omega(n^{5/3})$. Lemma 2.2 gives the query lower bound. $\square$

3 Finite-field residual transfer

Fix a finite field $\mathbb{F}_q$. All constants in this section are absolute and independent of q . We first work on the entire field-valued residual space, then restrict to the Boolean cube to invoke Section 2.

3.1 Characters and coordinate degree

Choose a nontrivial additive character $\chi : \mathbb{F}_q \rightarrow \mathbb{C}$, for example $\chi(a) = \exp(2\pi i \operatorname{Tr}_{\mathbb{F}_q/\mathbb{F}_\ell}(a)/\ell)$ when $q = \ell^s$ with ℓ prime (identifying the trace value with its standard integer representative in the exponential). The trace map is nonzero, hence the displayed character is nontrivial. For $u \neq 0$, multiplication by u permutes $\mathbb{F}_q$; character orthogonality therefore gives $\mathbb{E}_{a \in \mathbb{F}_q} \chi(ua) = \mathbf{1}[u = 0]$. For matrices put $\langle S, D \rangle = \sum_{i,j} S_{ij} D_{ij}$ and let $\operatorname{wt}(S)$ count nonzero entries. The characters on $\Omega_{n,q}$ are

$$(A, B, C) \mapsto \chi(\langle U, A \rangle + \langle V, B \rangle + \langle S, C \rangle).$$

Write $V_L = V_L(\Omega_{n,q})$ for the coordinate filtration already defined in Section 2.

Lemma 3.1 (Fourier description of coordinate degree). *V_L consists precisely of the real-valued functions whose Fourier coefficients vanish whenever $\operatorname{wt}(U) + \operatorname{wt}(V) + \operatorname{wt}(S) > L$.*

Proof. The one-coordinate identities

$$\mathbf{1}[z_t = a] = q^{-1} \sum_{u \in \mathbb{F}_q} \chi(u(z_t - a)), \quad \chi(uz_t) = \sum_{a \in \mathbb{F}_q} \chi(ua) \mathbf{1}[z_t = a]$$

show equality of the complex character and assignment spans at every support size. Taking real parts and applying Lemma 2.1 gives the stated description. $\square$

This degree is not finite-field algebraic degree. All polynomial lower bounds in this manuscript concern real functions and one-hot coordinates. No boundedness on invalid one-hot encodings is assumed or needed.

3.2 Random systematic codes

Set $\delta = 1/16$. Call B row-good if

$$\operatorname{wt}(v) + \operatorname{wt}(vB^\top) \geq \delta n \quad \text{for every nonzero row vector } v \in \mathbb{F}_q^n. \quad (5)$$

Call A column-good if

$$\operatorname{wt}(v) + \operatorname{wt}(A^\top v) \geq \delta n \quad \text{for every nonzero column vector } v \in \mathbb{F}_q^n. \quad (6)$$

These are distance conditions on systematic linear codes of length $2n$.

Lemma 3.2. *Put $\mathcal{H}(t) = -t \log t - (1-t) \log(1-t)$ and*

$$c_* = \frac{15}{16} \log 2 - 2\mathcal{H}(1/32) > 0, \quad \eta_n^F = e^{-c_* n}.$$

A uniform B fails (5) with probability at most η_n^F ; a uniform A fails (6) with the same bound.

Proof. For fixed nonzero v , the vector $vB^\top$ is uniform in $\mathbb{F}_q^n$, since its coordinates are independent nonzero linear functionals of the independent uniform rows of B . For each fixed $v \neq 0$ and each w , the probability of $vB^\top = w$ is therefore q^{-n} . Take a union bound over such pairs with $v \neq 0$ and concatenated weight at most $n/16$, and then enlarge their number by allowing $v = 0$. This gives

$$\begin{aligned} \Pr[B \text{ is not row-good}] &\leq q^{-n} \sum_{j=0}^{\lfloor n/16 \rfloor} \binom{2n}{j} (q-1)^j \\ &\leq \exp \left\{ -n \left(\log q - 2\mathcal{H}(1/32) - \frac{1}{16} \log(q-1) \right) \right\}. \end{aligned} \quad (7)$$

There are $\binom{2n}{j} (q-1)^j$ concatenated pairs of weight j . The first bound deliberately includes the non-failing boundary and the pairs with $v = 0$, so it remains an upper bound. The second uses the binomial entropy bound $\sum_{j \leq tm'} \binom{m'}{j} \leq e^{m' \mathcal{H}(t)}$ with $m' = 2n$ and $t = 1/32$, and $(q-1)^j \leq (q-1)^{n/16}$. Since $q \geq 2$ and $\log(q-1) \leq \log q$, $\log q - \frac{1}{16} \log(q-1) \geq \frac{15}{16} \log 2$, so this is at most $e^{-c_* n}$. Finally, $\mathcal{H}(t) \leq t \log(1/t) + t$ gives $c_* \geq (10 \log 2 - 1)/16 > 0$. The column case uses independent columns of A ; neither argument requires invertibility. $\square$

3.3 Transfer on field-valued residuals

For any real function p on $\Omega_{n,q}$, define

$$(\mathcal{R}p)(D) = \mathbb{E}_{A,B} p(A, B, AB - D), \quad D \in \mathbb{F}_q^{n \times n}, \quad (8)$$

where both factors are independent and uniform. For a residual frequency S , write $r(S)$ and $c(S)$ for its numbers of nonzero rows and columns. A function on residuals has *row Fourier degree* at most K if every frequency S in its Fourier support satisfies $r(S) \leq K$, and *column Fourier degree* at most K if every such S satisfies $c(S) \leq K$. Let $\mathcal{W}_K^{(q)}$ be the real vector space of real functions on residuals with both row and column Fourier degree at most K .

Theorem 3.3 (Uniform residual transfer). *For each integer $L \geq 0$, put*

$$K = \lfloor 16L/n \rfloor, \quad M_{n,K} = \sum_{j=0}^{\min(K,n)} \binom{n}{j} 2^j, \quad \zeta_{n,L} = \eta_n^F (1 + 2M_{n,K}).$$

There is a real linear map $\mathcal{H}_L : V_L \rightarrow \mathcal{W}_K^{(q)}$ such that

$$\|\mathcal{R}p - \mathcal{H}_L p\|_\infty \leq \zeta_{n,L} \|p\|_\infty \quad (p \in V_L). \quad (9)$$

Moreover, for every $a > 0$ there is $n_1(a)$ such that $\zeta_{n,L} \leq e^{-c_ n/2}$ for all $n \geq n_1(a)$ and all integers $0 \leq L \leq an^{5/3}$.*

Proof. Fix B and expand $p(A, B, \cdot)$ in the characters of (A, C) :

$$\begin{aligned} p(A, B, C) &= \sum_{U,S} a_{U,S}(B) \chi(\langle U, A \rangle + \langle S, C \rangle), \\ a_{U,S}(B) &= 0 \quad \text{if } \text{wt}(U) + \text{wt}(S) > L. \end{aligned} \quad (10)$$

Here the coefficients are normalized with respect to uniform measure:

$$a_{U,S}(B) = \mathbb{E}_{A,C} p(A, B, C) \chi(-\langle U, A \rangle - \langle S, C \rangle) = \sum_V \hat{p}(U, V, S) \chi(\langle V, B \rangle),$$

where $\hat{p}(U, V, S)$ denotes the Fourier coefficient of p on $\Omega_{n,q}$ at the frequency (U, V, S) . Since $p \in V_L$, every term of the last sum with $\text{wt}(U) + \text{wt}(S) > L$ vanishes, which justifies the vanishing condition in (10). The matrix identity $\langle S, AB \rangle = \langle SB^\top, A \rangle$ and orthogonality in the *unconditioned* uniform variable A give

$$g_B(D) := \mathbb{E}_A p(A, B, AB - D) = \sum_S a_{-SB^\top, S}(B) \chi(-\langle S, D \rangle). \quad (11)$$

By the vanishing condition in (10), a surviving residual frequency (one with a nonzero coefficient in (11)) satisfies

$$\text{wt}(S) + \text{wt}(SB^\top) \leq L. \quad (12)$$

If B is row-good, sum (5) over the nonzero rows of S . The resulting left side is exactly $\text{wt}(S) + \text{wt}(SB^\top)$, so it is at least $\delta nr(S)$. Since $r(S)$ is integral, (12) gives $r(S) \leq \lfloor 16L/n \rfloor = K$. The subnormalized average

$$g_{\text{row}}(D) = \mathbb{E}_B [\mathbf{1}[B \text{ row-good}] g_B(D)]$$

has row Fourier degree at most K . With $g = \mathcal{R}p$,

$$\|g - g_{\text{row}}\|_\infty \leq \eta_n^F \|p\|_\infty. \quad (13)$$

Indeed, the omitted set of B has probability at most η_n^F , and $|g_B(D)| \leq \|p\|_\infty$ uniformly in B, D .

Independently fix A and write

$$p(A, B, C) = \sum_{V, S} b_{V, S}(A) \chi(\langle V, B \rangle + \langle S, C \rangle), \quad b_{V, S}(A) = 0 \text{ if } \text{wt}(V) + \text{wt}(S) > L,$$

where the vanishing holds for the same reason as in (10). Since $\langle S, AB \rangle = \langle A^\top S, B \rangle$, the corresponding average is

$$\begin{aligned} g_{\text{col}}(D) &= \mathbb{E}_A [\mathbf{1}[A \text{ column-good}] \mathbb{E}_B p(A, B, AB - D)] \\ &= \mathbb{E}_A \left[\mathbf{1}[A \text{ column-good}] \sum_S b_{-A^\top S, S}(A) \chi(-\langle S, D \rangle) \right], \end{aligned}$$

where the inner B average is uniform and unconditioned. Every surviving frequency for a column-good A satisfies

$$\text{wt}(S) + \text{wt}(A^\top S) \leq L.$$

Summing (6) over the nonzero columns of S gives $\text{wt}(S) + \text{wt}(A^\top S) \geq \delta nc(S)$ and hence $c(S) \leq K$. Thus g_{col} has column Fourier degree at most K and

$$\|g - g_{\text{col}}\|_\infty \leq \eta_n^F \|p\|_\infty. \quad (14)$$

These are two separate, subnormalized averages. In each calculation the variable used for character orthogonality remains unconditioned; we have not conditioned both factors to be good.

For each row i of D , let E_i be uniform averaging over that entire row while holding the other entries fixed. These are commuting projections of L_∞ norm one. For $R \subseteq [n]$, define

$$Q_R = \prod_{i \in R} (I - E_i) \prod_{i \notin R} E_i, \quad P_K = \sum_{|R| \leq K} Q_R.$$

On a character $\chi(\langle S, D \rangle)$, E_i is the identity if row i of S is zero and is zero otherwise. Hence Q_R selects precisely row support R , and P_K selects row degree at most K . Also

$$\|P_K\|_{\infty \rightarrow \infty} \leq \sum_{|R| \leq K} 2^{|R|} = M_{n,K}. \quad (15)$$

This projection is not assumed to be an L_∞ contraction; its full $M_{n,K}$ cost is retained. No factor involving the row alphabet size q^n occurs. The projection is diagonal in the entrywise Fourier basis and only deletes frequencies. It therefore cannot introduce a frequency with more than K nonzero columns. Consequently $\mathcal{H}_{Lp} := P_K g_{\text{col}}$ has both required support restrictions. Since $P_K g_{\text{row}} = g_{\text{row}}$, the triangle inequality and (13)–(14) give

$$\begin{aligned} \|P_K g_{\text{col}} - g\|_\infty &\leq \|g_{\text{row}} - g\|_\infty + \|P_K(g_{\text{col}} - g_{\text{row}})\|_\infty \\ &\leq \eta_n^F \|p\|_\infty + M_{n,K}(2\eta_n^F \|p\|_\infty), \end{aligned}$$

which proves (9). The averages and row projections preserve real functions, proving the asserted real linearity.

For $L \leq an^{5/3}$, $K \leq 16an^{2/3} < n$ for all sufficiently large n . If $K \geq 1$, the crude estimate $M_{n,K} \leq (K+1)(2n)^K$ gives $\log M_{n,K} = O_a(n^{2/3} \log n) = o(n)$; if $K = 0$ the assertion is immediate. Consequently $\log(1 + 2M_{n,K}) \leq c_* n/2$ for all sufficiently large n , uniformly in this range of L . Multiplying by $e^{-c_* n}$ proves the stated error bound. $\square$

3.4 Completing the finite-field lower bound

Proof of Theorem 1.2. Choose A, B independently and uniformly over $\mathbb{F}_q^{n \times n}$, and restrict residual averaging and the operator of Theorem 3.3 to $\mathcal{D}_n$. A field character with at most K nonzero rows and columns depends only on the corresponding rectangle of residual entries. Its restriction to $\{0, 1\}^{n \times n}$ therefore lies in the complex span of the Boolean space $\mathcal{W}_K$. For a real-valued combination, taking real parts gives membership in the real space. No identification of the field character with a Boolean character is required; in particular, no condition on $\chi(1)$ is used.

Restriction cannot increase the supremum norm. Hence the two restricted operators satisfy Definition 2.3, uniformly in q . Every sampled triple has $AB - C = D$, so the required fiber condition also holds. Proposition 2.7 proves the coordinate-degree and query lower bounds. The matching upper bound is the stored-submatrix walk of [8]; its ring-independent query analysis is recalled in Section 5. $\square$

4 Integer lattice masking and residual transfer

We now use ordinary integer arithmetic and specify the finite boxes underlying Theorem 1.3. For a positive integer H , set

$$\begin{aligned} I_H &= [-2H, 2H] \cap \mathbb{Z}, & J_{n,H} &= [-4nH^2 - 1, 4nH^2 + 1] \cap \mathbb{Z}, \\ \mathcal{X}_{n,H} &= I_H^{n \times n} \times I_H^{n \times n} \times J_{n,H}^{n \times n}, & H_n &= 2^{100n}, & \mathcal{X}_n &= \mathcal{X}_{n,H_n}. \end{aligned}$$

We prove the following quantitative form of Theorem 1.3: there are absolute $c > 0$ and n_0 such that for every $n \geq n_0$ and integer $H \geq H_n$,

$$\widetilde{\deg}_{\text{bd},1/3}(f|_{\mathcal{X}_{n,H}}) \geq cn^{5/3}, \quad Q_{1/3}(f|_{\mathcal{X}_{n,H}}) = \Theta(n^{5/3}),$$

with constants independent of H . The choice $H = H_n$ gives the $O(n)$ -bit restriction in the headline theorem. The integer assertions of Theorem 1.4 use this same family: they hold for every $H \geq H_n$, with the factors sampled from $[-H, H]$ in its distributional assertion.

Random factors are sampled from the inner box $[-H, H]^{n \times n} \cap \mathbb{Z}^{n \times n}$. This distinction is essential: the function norm and boundedness are on the outer domain, which also contains the translated arguments.

Unlike finite-field character averaging, integer averaging is only approximately invariant under a residual shift. We first construct short integer lifts uniformly over all sparse observations, then correct each row operator before tensoring. The arithmetic exceptional set and the much smaller translation error are controlled separately.

4.1 Effective surjectivity and short integer lifts

For a rank- d lattice Λ in its Euclidean span, $\det \Lambda$ is its covolume and $\lambda_i(\Lambda)$ is the least radius of a centered Euclidean ball containing i linearly independent lattice vectors. Let B_2^d be the Euclidean unit ball. We use Minkowski's second theorem in its standard form

$$\prod_{i=1}^d \lambda_i(\Lambda) \leq \frac{2^d}{\text{vol}(B_2^d)} \det \Lambda \quad (16)$$

for a rank- d Euclidean lattice; see the general lattice statement in [2, Section 2, equation (12)]. An asymptotic version of the arithmetic step below also follows from the rectangular-unimodular density theorem [19, Corollary 17]. We instead give an elementary finite-box estimate, needed to control the entry bit length.

Lemma 4.1 (Effective surjectivity). *Let k, m, H be positive integers with $m > 2k$ and $H \geq \max\{k, 4m\}$. Let T be a $k \times m$ matrix with independent uniform integer entries in $[-H, H]$. Put $d = m - k$ and $W = 2H + 1$. Then*

$$\begin{aligned} \Pr[T\mathbb{Z}^m \neq \mathbb{Z}^k] &\leq 3 \cdot 2^{-d} + \mathcal{E}_{k,m}(H), \\ \mathcal{E}_{k,m}(H) &= \frac{k}{W} + \frac{2^{m+1}(4m)^d}{dH^d} + 2k \cdot 2^d W^{-(m-2k+1)}. \end{aligned} \quad (17)$$

Proof. For a prime p , a rank-deficient matrix over $\mathbb{F}_p$ has all its columns in one of at most $2p^{k-1}$ hyperplanes. For a fixed hyperplane, condition on all but one coordinate with nonzero normal coefficient. An interval of W integers puts mass at most $p^{-1} + W^{-1}$ on a specified residue. Thus

$$\Pr[\text{rank}(T \bmod p) < k] \leq 2p^{k-1}(p^{-1} + W^{-1})^m. \quad (18)$$

If $p \leq H/(2m)$, the right side is at most $2e^{1/4}p^{-d-1}$. Since $d \geq 2$,

$$\sum_{r=2}^{\infty} r^{-d-1} \leq 2^{-d}(1/2 + 1/d) \leq 2^{-d}.$$

The union over these small primes costs less than $3 \cdot 2^{-d}$.

For $H/(2m) < p \leq W$, (18) is at most $2^{m+1}p^{-d-1}$. With $P_0 = \lfloor H/(2m) \rfloor \geq H/(4m)$, their total contribution is at most

$$2^{m+1} \sum_{r > P_0} r^{-d-1} \leq \frac{2^{m+1}}{dP_0^d} \leq \frac{2^{m+1}(4m)^d}{dH^d}.$$

For $p > W$, first expose the square matrix S_0 formed by the first k columns. A nonzero polynomial of total degree k vanishes on at most a k/W fraction of a Cartesian grid with W choices per coordinate (the Schwartz–Zippel lemma [20]). Applying this grid root bound to the determinant gives $\Pr[\det S_0 = 0] \leq k/W$. Condition on the exposed matrix S_0 with $\Delta_0 = \det S_0 \neq 0$. By Hadamard’s inequality and $H \geq k$,

$$|\Delta_0| \leq (\sqrt{k}H)^k < W^{2k}.$$

Thus fewer than $2k$ primes greater than W divide Δ_0 . Any large prime causing rank failure must be among this list, which depends only on S_0 .

Fix such a prime and expose the remaining d columns. While their span together with S_0 is proper, the probability that a new column fails to increase its rank is at most $1/W$: one nonzero linear equation allows at most one of the $W < p$ possible pivot values. Define a failed step to be false once full rank has been reached. This bound then holds conditionally on every previous history. If final rank is still below k , at least $d - k + 1 = m - 2k + 1$ steps failed. Repeated conditional expectation and a union bound over their positions give probability at most $2^d W^{-(m-2k+1)}$. Union over the fewer than $2k$ candidate primes gives the final term of $\mathcal{E}_{k,m}(H)$.

Outside $\det S_0 = 0$, the image of T has finite index in $\mathbb{Z}^k$. It is a proper subgroup exactly when it fails to have full rank modulo some prime, by Smith normal form. This proves (17). $\square$

We will also use the elementary kernel-covolume identity: for every surjective integer map $T : \mathbb{Z}^m \rightarrow \mathbb{Z}^k$,

$$\det(\ker T \cap \mathbb{Z}^m) = \sqrt{\det(TT^\top)}. \quad (19)$$

Indeed, choose an integer right inverse U and an integer basis matrix Y of the kernel. The columns of $[Y \ U]$ form a basis of $\mathbb{Z}^m$. In orthonormal coordinates splitting the real kernel and its orthogonal complement, the two diagonal-block determinants have absolute values $\det(\ker T \cap \mathbb{Z}^m)$ and $1/\sqrt{\det(TT^\top)}$. Their product is one.

Lemma 4.2 (Euclidean primitive-vector estimate). *Let T and W be as in Lemma 4.1, and put $d = m - k$. For a primitive nonzero $v \in \mathbb{Z}^m$, write $r_v = \|v\|_2$. If x is uniform in $[-H, H]^m \cap \mathbb{Z}^m$, then*

$$\Pr[x \cdot v = 0] \leq \frac{\sqrt{2}}{Wr_v} \exp\left(\frac{(m-1)^2 r_v}{W}\right). \quad (20)$$

Consequently, for every real $R \geq \sqrt{m}$,

$$\begin{aligned} & \Pr[\exists 0 \neq v \in \ker T \cap \mathbb{Z}^m : \|v\|_2 \leq R] \\ & \leq 2^m \left(\frac{\sqrt{2}}{W}\right)^k \exp\left(\frac{k(m-1)^2 R}{W}\right) (1 + m^{-m/2} R^d). \end{aligned} \quad (21)$$

Proof. The lattice $\Lambda_v = v^\perp \cap \mathbb{Z}^m$ has covolume r_v by (19), since v is primitive. Choose j with $v_j \neq 0$. The $m-1$ independent kernel vectors $v_j e_i - v_i e_j$, $i \neq j$, have length at most r_v . Rounding coefficients in their parallelepiped shows that the covering radius of Λ_v is at most $\mu = (m-1)r_v/2$; they need not generate the full lattice.

The Euclidean Voronoi cell of the full lattice has volume r_v and radius at most μ . Its translates centered at lattice points in $[-H, H]^m$ have disjoint interiors and lie in $v^\perp \cap [-H - \mu, H + \mu]^m$. Ball’s

cube-slicing theorem [5, 18] bounds that section's volume by $\sqrt{2}(2H + (m-1)r_v)^{m-1}$. Dividing by $r_v W^m$ gives

$$\Pr[x \cdot v = 0] \leq \frac{\sqrt{2}}{W r_v} \left(\frac{2H + (m-1)r_v}{W} \right)^{m-1} \leq \frac{\sqrt{2}}{W r_v} \left(1 + \frac{(m-1)r_v}{W} \right)^{m-1}.$$

Now $1 + t \leq e^t$ proves (20).

Let $N(t)$ count nonzero integer vectors of Euclidean norm at most t . Disjoint unit cubes centered at such vectors fit in the ball of radius $t + \sqrt{m}/2$. The standard bound $\text{vol}(B_2^m) \leq (\sqrt{2\pi e}/\sqrt{m})^m$ gives

$$N(t) \leq 8^m \quad (1 \leq t \leq \sqrt{m}), \quad N(t) \leq 8^m m^{-m/2} t^m \quad (t \geq \sqrt{m}).$$

The two ranges use respectively $t + \sqrt{m}/2 \leq 3\sqrt{m}/2$ and $t + \sqrt{m}/2 \leq 3t/2$, together with $(3/2)\sqrt{2\pi e} < 8$. Partial summation, including the possible atom at norm one, yields

$$\begin{aligned} \sum_{0 < \|v\|_2 \leq R} \|v\|_2^{-k} &= N(R)R^{-k} + k \int_1^R N(t)t^{-k-1} dt \\ &\leq 8^m \left[1 + (1 + k/d)m^{-m/2}R^d \right] \leq 2 \cdot 8^m \left(1 + m^{-m/2}R^d \right). \end{aligned}$$

Here $k \leq d$. Every nonzero integer kernel vector has a primitive divisor in the kernel. Raise (20) to the k th power for the independent rows, then sum over primitive vectors, overcounting them by the displayed sum. This proves (21). $\square$

Lemma 4.3 (Quantitative short lifts). *Let n, k, m, H be integers satisfying $n \geq 16$, $1 \leq k \leq n/16$, $15n/16 \leq m \leq n$, and $H \geq 2^{20n}$. Let T be a $k \times m$ matrix with independent uniform integer entries in $[-H, H]$. Apart from an event of probability at most*

$$2^{6n}H^{-3/7} + 2n^2/H, \tag{22}$$

the following implication holds: if $T\mathbb{Z}^m = \mathbb{Z}^k$, then every $u \in \{-1, 0, 1\}^k$ has an integer lift with

$$Tw = u, \quad \|w\|_1 \leq 2^n n^{3/2} \sqrt{H}. \tag{23}$$

Proof. Put $W = 2H + 1$, $d = m - k \geq 14k$, and

$$\beta = \frac{k - 1/2}{d - 1}, \quad b = m^{m/(2(d-1))}, \quad R = \lfloor bH^\beta \rfloor.$$

The identity $k - \beta d = (d/2 - k)/(d - 1)$ and the bound $d \geq 14k$, together with the other hypotheses, give

$$\frac{1}{2(n-2)} \leq \beta < \frac{1}{14}, \quad k - \beta d > \frac{3}{7}, \quad \sqrt{m} \leq b \leq m, \quad R \geq bH^\beta/2 \geq \sqrt{m}.$$

For the last inequalities, $H^\beta \geq 2^{10}$ and $\lfloor x \rfloor \geq x/2$ for $x \geq 2$ justify the floor in R . Since $R \leq bH^\beta \leq nH^{1/14}$ and $n^4 \leq 2^n$ for $n \geq 16$, we also have

$$\frac{k(m-1)^2 R}{W} \leq \frac{n^4}{2H^{13/14}} \leq 1.$$

Since $m^{-m/2}b^d = b \leq n$, (21) bounds the short-vector probability by

$$2e \cdot 8^n 2^{n/2} (1+n) H^{-3/7} \leq 2^{6n} H^{-3/7}.$$

On integer surjectivity, $\Lambda = \ker T \cap \mathbb{Z}^m$ is a rank- d lattice. Equation (19) and Hadamard's inequality give $\det \Lambda \leq m^{k/2} H^k$.

Outside the short-vector event, the first successive minimum exceeds R . Minkowski's theorem (16) and $\text{vol}(B_2^d) \geq (2/\sqrt{d})^d$ yield

$$\lambda_d(\Lambda) \leq \frac{d^{d/2} m^{k/2} H^k}{R^{d-1}} \leq 2^{d-1} (d/m)^{d/2} \sqrt{H} \leq 2^n \sqrt{H}.$$

The middle inequality uses $R \geq bH^\beta/2$ and the exact cancellation $b^{d-1} = m^{m/2}$.

For rows t_i of T , put $r_i = \text{dist}(t_i, \text{span}\{t_j : j \neq i\})$. Condition on the other rows and choose a unit normal v to their span. Some $|v_j| \geq 1/\sqrt{m}$. The event $r_i < \sqrt{k}$ forces $|v \cdot t_i| < \sqrt{k}$, which, after fixing the other entries of t_i , allows at most $2\sqrt{km} + 1$ integer pivot values. Consequently

$$\Pr[\text{some } r_i < \sqrt{k}] \leq \frac{k(2\sqrt{km} + 1)}{W} \leq \frac{2n^2}{H}.$$

On the complementary event the rows are independent, and the Schur-complement identity

$$\text{tr}((TT^\top)^{-1}) = \sum_i r_i^{-2} \leq 1$$

implies $\sigma_{\min}(T) \geq 1$.

For $\|u\|_2 \leq \sqrt{k}$, the minimum-norm real solution has norm at most $\sqrt{k}$. Subtracting a lattice vector from any integer solution puts it within $d\lambda_d/2$ of that real solution: choose d independent kernel vectors of lengths at most λ_d and round coefficients in their parallelepiped. A basis of the full lattice is not required. Thus

$$\|w\|_1 \leq \sqrt{m}(\sqrt{k} + d\lambda_d/2) \leq n + 2^{n-1} n^{3/2} \sqrt{H} \leq 2^n n^{3/2} \sqrt{H}.$$

Combining the exceptional events proves the lemma. $\square$

For a fixed $w \in \mathbb{Z}^m$ and a uniform on $[-H, H]^m \cap \mathbb{Z}^m$,

$$d_{\text{TV}}(a, a + w) = 1 - \prod_j \left(1 - \frac{|w_j|}{2H + 1}\right)_+ \leq \frac{\|w\|_1}{2H + 1}. \quad (24)$$

Here d_{TV} is total-variation distance between probability laws, $d_{\text{TV}}(\mu, \nu) = \frac{1}{2} \sum_x |\mu(x) - \nu(x)|$, and $(t)_+ = \max\{t, 0\}$. The product counts the fraction of box points in the overlap of the two translates; the inequality follows from $1 - \prod_j (1 - a_j) \leq \sum_j a_j$ for $a_j \in [0, 1]$ when all $|w_j| \leq 2H + 1$. If a displacement exceeds $2H + 1$, the right side of (24) is already at least one. Thus the lifts hide bounded shifts at equal factor widths.

4.2 Uniformity over sparse observations

Put $\delta = 1/16$ and let $\mathcal{H}$ be the entropy function of Lemma 3.2, so that

$$c_* = (1 - \delta) \log 2 - 2\mathcal{H}(\delta/2) = \frac{15}{16} \log 2 - 2\mathcal{H}(1/32)$$

is the constant of that lemma; numerically $c_* > 0.3717$. Define

$$\eta_n^Z = 3e^{-c_* n}, \quad H_n = 2^{100n}.$$

A random factor B has independent uniform entries in $[-H, H] \cap \mathbb{Z}$.

Lemma 4.4 (Explicit good factors). *For every $n \geq 16$ and integer $H \geq H_n$, outside a set of B of probability at most $\eta_n^Z + 2^{-33n}$, every map*

$$T_{I,J} = B[I^c, J]^\top : \mathbb{Z}^{n-u} \longrightarrow \mathbb{Z}^v, \quad u = |I|, \quad v = |J| \geq 1, \quad u + v \leq \delta n,$$

is onto and every $z \in \{-1, 0, 1\}^v$ has a preimage with ℓ_1 norm at most $\rho_n H$, where $\rho_n = 2^{-48n}$. Here $I^c = [n] \setminus I$. The corresponding statement holds for an independent factor A with the same entry law and maps $A[J, I^c]$, with the same probability bound.

Proof. Each pattern meets the hypotheses of Lemmas 4.1 and 4.3, with $k = v$, $m = n - u$, and $d_0 = n - u - v \geq 15n/16$. Indeed $1 \leq k = v \leq \delta n = n/16$ and $m = n - u \geq n - \delta n + v \geq 15n/16$, so $m > 2k$; moreover $H \geq H_n = 2^{100n} \geq \max\{k, 4m, 2^{20n}\}$ and $n \geq 16$. The leading arithmetic terms sum to at most

$$3 \cdot 2^{-n} \sum_{r \leq \delta n} \binom{2n}{r} 2^r \leq 3 \exp\{-n[(1 - \delta) \log 2 - 2\mathcal{H}(\delta/2)]\} = \eta_n^Z.$$

There are at most 4^n patterns. Their geometric remainders sum to

$$2^{8n} H^{-3/7} + 2^{2n+1} n^2 / H \leq 2^{-34n}.$$

Indeed their binary logarithms are at most $-(244/7)n$ and $-97n$. For the three arithmetic remainders in (17), including the factor 4^n , binary logarithms are bounded respectively by

$$-97n, \quad 3n + 1 - \frac{2985}{32} n^2, \quad 3n + 1 + \log_2 n - 100n(7n/8 + 1).$$

Here $\log_2(4n) \leq n/2$, $d_0 \geq 15n/16$, and $m - 2k + 1 = n - u - 2v + 1 \geq 7n/8 + 1$. The first arithmetic error term is at most 2^{-97n} and the other two are each at most 2^{-98n} . Their sum is therefore at most 2^{-96n} ; together with the geometric error this is below 2^{-33n} . Finally,

$$\frac{2^n n^{3/2}}{\sqrt{H}} \leq 2^{-49n} n^{3/2} \leq 2^{-48n},$$

so Lemma 4.3 supplies every required lift. All bounds decrease with H . $\square$

4.3 Row-local correction and residual transfer

Residuals in this section belong to the finite Boolean cube $\mathcal{D}_n = \{0, 1\}^{n \times n}$. This is a set of integer residuals; it does not change multiplication to Boolean or mod-two multiplication. Use the Boolean characters and space $\mathcal{W}_K$ of Section 2, and put $V_L = V_L(\mathcal{X}_{n,H})$. A function has row Fourier degree at most K if every character in its Fourier expansion meets at most K rows; column Fourier degree is analogous. Thus $\mathcal{W}_K$ is the intersection of these subspaces. Operator norms in this subsection are induced by the supremum norms on the full finite domains. Define

$$(\mathcal{R}_{HP})(D) = \mathbb{E}_{A,B} p(A, B, AB - D),$$

where A, B are independent inner-box factors of width H .

Theorem 4.5 (Uniform integer transfer). *For every fixed $a > 0$, there is an $n_0(a)$ such that, for every $n \geq n_0(a)$ and integer $H \geq H_n$, simultaneously for integers $0 \leq L \leq an^{5/3}$ there is a real linear map $\mathcal{H}_{H,L} : V_L \rightarrow \mathcal{W}_K$, $K = \lfloor 16L/n \rfloor$, satisfying*

$$\|\mathcal{R}_{HP} - \mathcal{H}_{H,L} p\|_\infty \leq e^{-c_* n/2} \|p\|_\infty. \quad (25)$$

The box threshold $H_n = 2^{100n}$ is independent of p , L , and a .

Proof. We first construct a corrected averaging operator for one row. All spaces below are finite and carry the supremum norm. Write

$$\mathcal{Z} = I_H^n \times J_{n,H}^n, \quad \mathcal{D} = \{0, 1\}^n,$$

and, for a fixed inner-box factor B , define

$$(T_B F)(d) = \mathbb{E}_{x \in [-H, H]^n \cap \mathbb{Z}^n} F(x, xB - d), \quad (S_B F)(d) = (T_B F)(0),$$

where x is a row vector and xB is its product with B . Both operators map functions on $\mathcal{Z}$ to functions on $\mathcal{D}$ and have norm one; S_B has constant output.

Suppose B is good in Lemma 4.4. If F depends only on the coordinates I of its first argument and the coordinates J of its second argument, with $|I| + |J| < n/16$, then

$$\|(T_B - S_B)F\|_\infty \leq \rho_n \|F\|_\infty. \quad (26)$$

Indeed, for each d , choose an integer w zero on I such that $wB[:, J] = -d_J$ and $\|w\|_1 \leq \rho_n H$. If J is empty, take $w = 0$. The observed arguments of F at $(x, xB - d)$ and $(x + w, (x + w)B)$ agree. Since $\rho_n \leq 1$, we have $|x_j + w_j| \leq 2H$ and $|((x + w)B)_j| \leq 2nH^2$; also $|(xB - d)_j| \leq nH^2 + 1$. Hence both arguments lie in $\mathcal{Z}$, and (24) gives $d_{TV}(x, x + w) \leq \rho_n/2$. Apply the elementary inequality $|\mathbb{E}_\mu G - \mathbb{E}_\nu G| \leq 2d_{TV}(\mu, \nu) \|G\|_\infty$, valid for bounded G and probability measures μ, ν , to $G(y) = F(y, yB)$ with μ, ν the laws of x and $x + w$; on the union of their supports $|G| \leq \|F\|_\infty$. This proves (26). The translation w does not depend on x or on the values of F .

Put $r = \lceil n/16 \rceil$. For each of the $2n$ coordinates of $\mathcal{Z}$, let Z_t set that coordinate to zero. Define the anchored local projection

$$\Pi_S = \prod_{t \in S} (I - Z_t) \prod_{t \notin S} Z_t, \quad \Pi = \sum_{|S| < r} \Pi_S, \quad d_n = \sum_{s=0}^{r-1} \binom{2n}{s} 2^s \leq 3^{2n}.$$

These operators act on all functions on $\mathcal{Z}$, not merely on a low-degree subspace. The operators Z_t commute and have norm one. Their products satisfy $\sum_S \Pi_S = I$. If F depends on a coordinate set U , then $\Pi_S F = 0$ whenever $S \not\subseteq U$. Thus $\Pi_S F$ depends only on S and has norm at most $2^{|S|} \|F\|_\infty$; Π is the identity on functions of coordinate degree less than r on $\mathcal{Z}$. Since $|S| < r$ implies $|S| < n/16$, (26) gives

$$\|(T_B - S_B)\Pi\| \leq \rho_n d_n =: \epsilon_n.$$

Define the signed, not necessarily positive, operator

$$\tilde{T}_B = T_B - (T_B - S_B)\Pi.$$

Then

$$\|\tilde{T}_B - T_B\| \leq \epsilon_n, \quad \|\tilde{T}_B\| \leq 1 + \epsilon_n, \quad \tilde{T}_B F = S_B F \quad (F \in V_{r-1}(\mathcal{Z})). \quad (27)$$

For fixed B , regard $p_B = p(A, B, C)$ as a function on $\mathcal{Z}^n$. It has norm at most $\|p\|_\infty$ and coordinate degree at most L . Conditional residual averaging is exactly $T_B^{\otimes n} p_B$. The corrected image $\tilde{T}_B^{\otimes n} p_B$ has row Fourier degree at most $\lfloor L/r \rfloor \leq K$. To verify this, expand algebraically in assignment indicators on at most L coordinates. Such an indicator factors across rows. If it specifies $s_i < r$ coordinates in row i , (27) makes that output row constant. Since $\sum_i s_i \leq L$, at most $\lfloor L/r \rfloor$ rows remain. This expansion identifies the output subspace only; no bound on the expansion's coefficients is used.

Supremum operator norms on finite spaces are maximum absolute row sums and therefore multiply under tensor products, even for signed operators. Telescoping (27) therefore gives

$$\|\tilde{T}_B^{\otimes n} - T_B^{\otimes n}\| \leq \epsilon_n \sum_{j=0}^{n-1} (1 + \epsilon_n)^j = (1 + \epsilon_n)^n - 1. \quad (28)$$

Set

$$q_{\text{row}}(D) = \mathbb{E}_B \mathbf{1}[B \text{ good}] (\tilde{T}_B^{\otimes n} p_B)(D).$$

The integrand is defined to be zero for bad B . This average is linear in p and has row Fourier degree at most K . With $g = \mathcal{R}_H p$, discarding bad B on the original norm-one averaging operator costs only their probability. Hence

$$\|g - q_{\text{row}}\|_{\infty} \leq (\eta_n^Z + 2^{-33n} + (1 + \epsilon_n)^n - 1) \|p\|_{\infty}. \quad (29)$$

Here $\epsilon_n \leq 3^{2n} 2^{-48n} \leq 2^{-44n}$. For $n \geq 16$,

$$(1 + \epsilon_n)^n - 1 \leq e^{n\epsilon_n} - 1 \leq 2n\epsilon_n \leq 2^{-43n}.$$

Since $c_* < \log 2$, the coefficient in (29) is at most $2\eta_n^Z$. Apply the same construction separately to $(B^\top, A^\top, C^\top)$, fixing good A . This gives q_{col} of column Fourier degree at most K with the same bound. Neither construction conditions both factors to be good.

Define the row projection P_K on functions of Boolean residuals as in the proof of Theorem 3.3, with E_i now the uniform average over row i of $D \in \{0, 1\}^{n \times n}$: on a Boolean character χ_S , E_i is the identity if row i of S is empty and zero otherwise, so $P_K = \sum_{|R| \leq K} \prod_{i \in R} (I - E_i) \prod_{i \notin R} E_i$ retains exactly the characters meeting at most K rows. It has norm at most $M_{n,K}$ by the same count, fixes q_{row} , and preserves column degree because it only deletes characters. Thus $\mathcal{H}_{H,L} p = P_K q_{\text{col}} \in \mathcal{W}_K$, and the same triangle estimate, using both one-sided errors $2\eta_n^Z \|p\|_{\infty}$, gives

$$\|g - \mathcal{H}_{H,L} p\|_{\infty} \leq 2\eta_n^Z (1 + 2M_{n,K}) \|p\|_{\infty}. \quad (30)$$

As there, $\log M_{n,K} = O_a(n^{2/3} \log n) = o(n)$ uniformly for $L \leq an^{5/3}$, including $K = 0$. Since $\eta_n^Z = 3e^{-c_* n}$, this proves (25) after increasing $n_0(a)$, which we also take to be at least 16 so that Lemma 4.4 applies. All operations defining $\mathcal{H}_{H,L}$ are real linear. $\square$

4.4 Completing the integer lower bound

Proof of Theorem 1.3 and its quantitative form. For each Boolean residual D , let μ_D be the law of $(A, B, AB - D)$ with independent inner-box factors. These triples belong to $\mathcal{X}_{n,H}$ and satisfy $f = \mathbf{1}[D = 0]$. Theorem 4.5 verifies Definition 2.3, uniformly over $H \geq H_n$. Proposition 2.7 supplies the degree and query lower bounds; the stored-submatrix verifier supplies the matching upper bound. At $H = H_n$, every entry in the outer domain has magnitude at most $4n 2^{200n} + 1$, hence has $O(n)$ bits. The oracle returns that complete exact value, so this statement is not a bit-query lower bound. $\square$

5 Optimal query complexity over unital rings

We combine the arithmetic branches by transferring finite restrictions to R , which may be infinite, noncommutative, or have zero divisors. This completes the query-complexity assertion of Theorem 1.1 using only the polynomial lower bounds of the preceding sections. The adversary assertion is deferred to Section 6.6.

Proof of Theorem 1.1: query complexity. Characteristic-zero lower bound. If $\text{char } R = 0$, the map $\iota : \mathbb{Z} \rightarrow R$ given by $\iota(a) = a1_R$ is an injective ring homomorphism with central image. Apply it to every entry of an integer instance. Then

$$\iota(A)\iota(B) - \iota(C) = \iota(AB - C),$$

so verification is preserved exactly. In particular, the finite hard restriction from the lower-bound assertion of Theorem 1.3 embeds in R without changing its matrix dimension.

Positive-characteristic lower bound. Let $\text{char } R = m \geq 2$, choose a prime $p \mid m$, and set

$$e = (m/p)1_R.$$

The additive order of 1_R is m , so e has additive order exactly p . For matrices A, B, C over $\mathbb{F}_p$, let $\hat{A}, \hat{B}, \hat{C}$ be their entrywise integer representatives in $\{0, \dots, p-1\}$. Define

$$A'_{ij} = \hat{A}_{ij}1_R, \quad B'_{ij} = \hat{B}_{ij}e, \quad C'_{ij} = \hat{C}_{ij}e. \quad (31)$$

Distributivity and multiplication by integer multiples of the identity give

$$(A'B' - C')_{ij} = \left(\sum_k \hat{A}_{ik} \hat{B}_{kj} - \hat{C}_{ij} \right) e. \quad (32)$$

Since an integer multiple te vanishes exactly when $p \mid t$, $A'B' = C'$ in R if and only if $AB = C$ in $\mathbb{F}_p$. We therefore apply the finite-prime-field query lower bound of Theorem 1.2. No embedded field in R is required. The encoding is intentionally asymmetric: e need not be idempotent, so multiplying both factors by e would not give (32).

Oracle preservation. Each transformed entry depends on one source entry through a fixed map. One can simulate its value oracle by querying that entry into an ancilla, computing the encoded answer reversibly, and uncomputing the ancilla, using a constant number of source queries.

Upper bound. Use the stored-submatrix variant *Verify Full* of Buhrman and Špalek [8, Sections 4.1 and 6.1]. More generally, consider $A \in R^{t \times k}$, $B \in R^{k \times t}$, and $C \in R^{t \times t}$ with $1 \leq t \leq 2k$. For $t \geq 8$, choose independent uniform r -subsets $U, V \subseteq [t]$, where $1 \leq r \leq t/2$, and cache

$$A[U, :], \quad B[:, V], \quad C[U, V].$$

Preparing the subset superposition and these data costs $S = O(kr + r^2) = O(kr)$ queries, using $r \leq t/2 \leq k$. The walk is the lazy product chain on $\binom{[t]}{r} \times \binom{[t]}{r}$: with probability $1/2$ stay put, otherwise choose one of the two coordinates uniformly and move it by one Johnson step. Each move replaces one selected row or column; updating the cache and uncomputing removed entries costs $U_{\text{qry}} = O(k + r) = O(k)$ queries. The Johnson graph $J(t, r)$ has normalized spectral gap $t/(r(t-r))$ and all its normalized eigenvalues are at least $-1/(t-r)$, so for $r \leq t/2$ the lazy product chain is reversible and ergodic with eigenvalue gap $\delta = \Theta(1/r)$ in the sense of [17]. A state is marked when $A[U, :]B[:, V] \neq C[U, V]$. This exact check uses only cached values and ordered sums, with no division or random scalar compression; the integral-domain hypothesis is therefore unnecessary, and the conditional phase flip is computed reversibly from the cache without queries. If $AB \neq C$, a fixed wrong entry is captured with probability $\varepsilon \geq (r/t)^2$; if $AB = C$, no state is marked. The quantum-walk bound [17, Theorem 3], with checking cost zero, gives bounded-error query cost

$$O\left(S + \frac{U_{\text{qry}}}{\sqrt{\delta\varepsilon}}\right) = O\left(kr + \frac{kt}{\sqrt{r}}\right). \quad (33)$$

Taking $r = \lfloor t^{2/3} \rfloor$, which satisfies $1 \leq r \leq t/2$ for $t \geq 8$, gives $O(kt^{2/3})$ queries. For $t < 8$, reading all $2kt + t^2$ entries costs $O(k)$ and suffices. Setting $t = k = n$ proves the $O(n^{5/3})$ upper bound. This is a query bound only: the exact cached checks may use substantially more arithmetic time and workspace. $\square$

In particular, Theorem 1.1 applies to every finite or infinite field, with no averaging over an infinite field required.

Why the hypotheses matter. Without an identity, the universal exponent can fail: give a nontrivial abelian group the multiplication $xy = 0$. Then verification reduces to testing $C = 0$ and has query complexity $\Theta(n)$, by quantum search over its n^2 entries and the matching lower bound on a two-element entry restriction [6]. The zero ring has only YES inputs and needs no queries. Thus both hypotheses in Theorem 1.1 matter.

6 Shared distributional and adversary lower bounds

With the universal-ring query bound established, we prove Theorem 1.4 for both arithmetic settings at once, and then complete the adversary assertion of Theorem 1.1 in Section 6.6. Let Ω be either $\Omega_{n,q}$ or $\mathcal{X}_{n,H}$, and $f = \mathbf{1}[AB = C]$ on that finite product domain. Use the corresponding Boolean residual averaging map and transfer operator established in Sections 3 and 4. The common arithmetic argument uses only Definition 2.3 and the corresponding averaging distributions. In particular, its constants are uniform in q and in $H \geq H_n$, respectively.

6.1 A quantitative collision test

Lemma 6.1 (A quantitative collision test). *Suppose $h \in \mathcal{W}_K$, $0 \leq h \leq 1$, and*

$$\alpha = h(0), \quad \beta = \frac{1}{n^2} \sum_{i,j} h(E_{ij}), \quad \alpha - \beta \geq 1/32.$$

Then $K \geq c_0 n^{2/3}$ for all sufficiently large n , where $c_0 > 0$ is an absolute constant.

Proof. Set $N = 32n^2$ and $M = 48n^2$. Given a source map $u : [N] \rightarrow [M]$, choose $2n$ distinct domain positions uniformly in order, put the first n into a list I and the last n into a list J , and set

$$\Psi(u) = \mathbb{E}_{I,J} h(D(u|_I, u|_J)).$$

For each choice the polynomial degree is at most $2K$ by Lemma 2.5: each sampled label is replaced by the value of u at its fixed sampled position. This substitution sends each one-hot label indicator to a single one-hot source indicator. The sampling law is independent of u , so its average has the same degree bound. Equal *values* at different positions do not identify the source coordinates. Also $0 \leq \Psi \leq 1$ for every source map, whether or not it satisfies the collision promise. If u is injective, $\Psi(u) = \alpha$.

Now fix any exactly two-to-one map u . Its equal-value pairs form a perfect matching on the N domain positions. Let Z count matched pairs with one endpoint in each sampled list. Equivalently, Z is the number of nonzero entries of $D(u|_I, u|_J)$. Writing Z as the sum of its n^2 cross-pair indicators gives

$$\mu := \mathbb{E}Z = \frac{n^2}{N-1}, \quad \mu_2 := \mathbb{E}Z(Z-1) = \frac{n^2(n-1)^2}{(N-1)(N-3)}. \quad (34)$$

For the first identity, each of the n^2 cross pairs is a matching edge with probability $1/(N-1)$. For the second, two matching edges cannot share an endpoint. There are $n^2(n-1)^2$ ordered pairs of disjoint cross edges, and the probability that both belong to a uniform perfect matching is $1/((N-1)(N-3))$. Thus

$$\mu \geq 1/32, \quad \frac{\mu_2}{\mu} = \frac{(n-1)^2}{32n^2-3} \leq 1/32. \quad (35)$$

Put $P_1 = \Pr[Z=1]$ and $P_{\geq 2} = \Pr[Z \geq 2]$. Since $z \leq z(z-1)$ for integers $z \geq 2$,

$$P_1 \geq \mu - \mu_2, \quad P_{\geq 2} \leq \mu_2/2.$$

On $Z=0$ the residual is zero. Conditional on $Z=1$, the sole nonzero position is uniform in $[n]^2$: independent permutations of the two list orders act transitively on these positions and preserve the sampling law and the event. Hence the conditional average of h is β . On $Z \geq 2$ it is some $w \in [0,1]$ (take any such w if this event is null). It follows that

$$\begin{aligned} \alpha - \Psi(u) &= P_1(\alpha - \beta) + P_{\geq 2}(\alpha - w) \\ &\geq (\mu - \mu_2)(\alpha - \beta) - \mu_2/2 \\ &\geq \mu \left(\frac{31}{32} \frac{1}{32} - \frac{1}{64} \right) \geq \frac{15}{32768} =: \gamma. \end{aligned} \quad (36)$$

To amplify, put $\theta = \alpha - \gamma/2$ and fix $r = \lceil 2\gamma^{-2} \log 6 \rceil$. The polynomial

$$A_\alpha(t) = \Pr[\text{Bin}(r, t)/r \geq \theta]$$

is the sum of the binomial terms with $j \geq \lceil r\theta \rceil$. The hypothesis gives $\alpha \geq 1/32$, while $\gamma < 1/32$, so $0 < \theta < 1$. Hence A_α has degree at most r and lies in $[0,1]$ throughout $[0,1]$. Hoeffding's inequality [13] gives $A_\alpha(\alpha) \geq 5/6$ and $A_\alpha(t) \leq 1/6$ for $0 \leq t \leq \alpha - \gamma$, since both tail events are at distance at least $\gamma/2$ from the mean. Thus $A_\alpha(\Psi)$ is a collision approximant of degree at most $2rK$, globally bounded on all source maps. The imported lower bound yields $2rK \geq c_{\text{col}}N^{1/3}$. Taking $c_0 = c_{\text{col}}32^{1/3}/(2r)$ proves the lemma; r is an absolute constant independent of α and h . $\square$

6.2 The uniform seed and average-case hardness

Let $\mathcal{A}$ be $\mathbb{F}_q^{n \times n}$ in the finite-field case and $([-H, H] \cap \mathbb{Z})^{n \times n}$ in the integer case. Let P_{yes} be the law of (A, B, AB) for independent uniform $A, B \in \mathcal{A}$, and let P_{no} be the law of $(A, B, AB - E_{ij})$ with (i, j) independently uniform. Define

$$\lambda_0 = \frac{1}{2}(P_{\text{yes}} - P_{\text{no}}). \quad (37)$$

For a signed measure ρ on Ω , write $\|\rho\|_1 = \sum_{z \in \Omega} |\rho(z)|$ and $\rho(p) = \sum_z \rho(z)p(z)$. Explicitly,

$$\lambda_0(A, B, C) = \begin{cases} (2|\mathcal{A}|^2)^{-1}, & (A, B) \in \mathcal{A}^2, C = AB, \\ -(2n^2|\mathcal{A}|^2)^{-1}, & (A, B) \in \mathcal{A}^2, AB - C = E_{ij} \text{ for some } i, j, \\ 0, & \text{otherwise.} \end{cases} \quad (38)$$

The two supports are disjoint, so $\|\lambda_0\|_1 = 1$, $\lambda_0(1) = 0$, and $\lambda_0(f) = 1/2$. Although the integer seed is supported on inner-box factors, all function norms and all LP variables below use the full outer domain $\mathcal{X}_{n,H}$.

Theorem 6.2 (Restricted norm bound). *There are $a > 0$ and n_0 such that, for all $n \geq n_0$ and integers $0 \leq L \leq an^{5/3}$,*

$$\sup\{|\lambda_0(p)| : p \in V_L, \|p\|_\infty \leq 1\} \leq 1/16. \quad (39)$$

Proof. Choose $a = \min\{1, c_0/32\}$, where c_0 is from Lemma 6.1. Suppose (39) fails. Replace p by $-p$ if necessary, so that $\lambda_0(p) > 1/16$, and put $p_+ = (1 + p)/2$. Then $0 \leq p_+ \leq 1$ on all triples. Its residual average $g = \mathcal{R}p_+$ satisfies the exact identity

$$g(0) - \frac{1}{n^2} \sum_{i,j} g(E_{ij}) = \lambda_0(p) > 1/16.$$

Indeed, the left side is $\mathbb{E}_{P_{\text{yes}}}p_+ - \mathbb{E}_{P_{\text{no}}}p_+ = (\mathbb{E}_{P_{\text{yes}}}p - \mathbb{E}_{P_{\text{no}}}p)/2$. Apply Corollary 2.4, with $K = \lfloor 16L/n \rfloor$. Uniformly over the indicated range of L , $\xi_{n,L} \leq 1/100$ for all sufficiently large n . The resulting $h \in \mathcal{W}_K$ is bounded in $[0, 1]$ and obeys

$$h(0) - \frac{1}{n^2} \sum_{i,j} h(E_{ij}) \geq \frac{1/16 - 2\xi_{n,L}}{1 + 2\xi_{n,L}} \geq \frac{1}{24} > \frac{1}{32}.$$

Lemma 6.1 forces $K \geq c_0 n^{2/3}$. On the other hand, $K \leq 16an^{2/3} \leq (c_0/2)n^{2/3}$, a contradiction. Increase n_0 to cover the large- n requirements of both lemmas. $\square$

Corollary 6.3 (Average-case hardness). *The distributional assertion of Theorem 1.4 holds.*

Proof. Let p be the probability that the algorithm outputs *equal*, and set $a_1 = \mathbb{E}_{P_{\text{yes}}}p$ and $b_1 = \mathbb{E}_{P_{\text{no}}}p$. Under the balanced distribution its success probability is $(a_1 + 1 - b_1)/2$, so success at least $2/3$ gives $a_1 - b_1 \geq 1/3$. The acceptance function is bounded on all inputs and has degree at most twice the worst-case query count T , by Lemma 2.2, including its finite-history argument for adaptive algorithms. Thus $s = 2p - 1 \in V_{2T}$ has $\|s\|_\infty \leq 1$ and

$$\lambda_0(s) = a_1 - b_1 \geq 1/3,$$

contradicting Theorem 6.2 if $2T \leq an^{5/3}$. This proves $T = \Omega(n^{5/3})$ without amplifying average success pointwise. $\square$

6.3 An exact finite dual witness

We use the assignment notation of Section 2; the empty assignment is included throughout.

For $0 \leq \ell \leq 3n^2$, consider the finite linear program

$$\begin{aligned} \tau_\ell = \min_{u_z, v_z \geq 0} \quad & \sum_{z \in \Omega} (u_z + v_z) \\ \text{subject to} \quad & \sum_{z \sim \alpha} (u_z - v_z) = \sum_{z \sim \alpha} \lambda_0(z) \quad \text{for every } |\alpha| \leq \ell. \end{aligned} \quad (40)$$

Lemma 6.4 (Exact norm extension). *The optimum is attained, and*

$$\tau_\ell = \sup_{p \in V_\ell, \|p\|_\infty \leq 1} |\lambda_0(p)|. \quad (41)$$

Moreover $\tau_0 = 0$, $\tau_{3n^2} = 1$, and τ_ℓ is nondecreasing.

Proof. The Jordan parts of λ_0 give a feasible point of value one; the feasible objective sublevel set at one is compact, so the optimum exists. With free dual variables y_α , put $p(z) = \sum_{|\alpha| \leq \ell} y_\alpha \mathbf{1}[z \sim \alpha]$. The two dual inequalities at z are precisely $-1 \leq p(z) \leq 1$, and the objective is $\lambda_0(p)$. Lemma 2.1 and finite LP strong duality give the supremum without the absolute value. The feasible set of p is symmetric under $p \mapsto -p$, which gives (41). At degree zero the only constraint is total mass; it is satisfied by $u = v = 0$ because $\lambda_0(1) = 0$. At full degree, the constraint for the complete assignment associated with each input z forces $u_z - v_z = \lambda_0(z)$, so every feasible objective is at least $\|\lambda_0\|_1 = 1$; the Jordan decomposition attains one. Adding constraints proves monotonicity. $\square$

Define the integer

$$L_n = \max\{\ell \in \{0, \dots, 3n^2\} : \tau_\ell \leq 1/16\}. \quad (42)$$

The dependence of L_n on the finite domain Ω is suppressed in the notation. It is well-defined and smaller than $3n^2$. Theorem 6.2 implies $L_n \geq \lfloor an^{5/3} \rfloor$ for all sufficiently large n . Thus the recipe does not need the numerical constant hidden in the imported collision lower bound as an input.

Choose an optimum of (40) for $\ell = L_n$, set $\sigma = u - v$, and define

$$\lambda = \lambda_0 - \sigma, \quad \Lambda = \|\lambda\|_1, \quad \phi = \lambda/\Lambda. \quad (43)$$

Proposition 6.5. *The signed measure ϕ satisfies*

$$\|\phi\|_1 = 1, \quad \phi(p) = 0 \ (p \in V_{L_n}), \quad \phi(f) \geq 7/17.$$

In particular $\phi(1) = 0$, and its positive and negative parts both have total mass $1/2$.

Proof. The LP gives $\|\sigma\|_1 \leq 1/16$. Its constraints say that σ and λ_0 agree on every assignment indicator of size at most L_n ; Lemma 2.1 extends this equality to all of V_{L_n} . Thus λ annihilates that space, and

$$\Lambda \leq 17/16, \quad \lambda(f) \geq 1/2 - 1/16 = 7/16.$$

In particular $\Lambda > 0$, and normalization gives the correlation bound. The empty-assignment constraint gives total mass zero, from which the final assertion follows. $\square$

6.4 The negative-weight adversary matrix

We specify the adversary, check the assignment-isometry conversion [7], and deduce the general-input adversary bound.

For disjoint sets X of YES inputs and Y of NO inputs, the rectangular negative-weight adversary value is

$$\text{Adv}^\pm(X, Y) = \sup_{\Gamma} \{ \|\Gamma\| : \|\Gamma \circ \Delta_t\| \leq 1 \text{ for every } t \}, \quad (\Delta_t)_{x,y} = \mathbf{1}[x_t \neq y_t].$$

Here Γ is real, has rows X and columns Y , $\circ$ is the entrywise product, and norms are Euclidean operator norms. For a total function we write $\text{Adv}^\pm(f) = \text{Adv}^\pm(f^{-1}(1), f^{-1}(0))$. The general adversary theorem gives bounded-error query complexity $\Omega(\text{Adv}^\pm(X, Y))$ for partial functions over finite input alphabets [12, 16]; the nonbinary filtering convention used below is also spelled out in [7, Proposition 4]. This agrees with the symmetric formulation by putting Γ and $\Gamma^\top$ in opposite off-diagonal blocks; the operator norm and all filtered norms are unchanged. The lower-bound argument also allows arbitrary Hilbert-space workspace: its progress function uses only finitely many input-pair inner products, and its per-query estimate uses $\|M \otimes I_{\mathcal{H}}\| = \|M\|$ for finite matrices M .

Together with the finite-label simulation in Section 1.1, this justifies its use on finite restrictions of an infinite ring.

Set $\pi_+ = 2\phi^+$ and $\pi_- = 2\phi^-$. These are probability measures on Ω . For every assignment α of size at most L_n , the identity $\phi(\mathbf{1}[\cdot \sim \alpha]) = 0$ gives $\pi_+(\{z : z \sim \alpha\}) = \pi_-(\{z : z \sim \alpha\})$. Put $m = \lfloor L_n/2 \rfloor$; assume n is large enough that $m \geq 1$. For $0 \leq k \leq m$ define matrices with rows indexed by $z \in \Omega$ and columns by assignments of size at most k :

$$(V_k^\pm)_{z,\alpha} = \sqrt{\pi_\pm(z)} \mathbf{1}[z \sim \alpha]. \quad (44)$$

Products of two assignment indicators vanish when inconsistent and otherwise are assignment indicators of size at most $2k \leq L_n$. Consequently

$$G_k = (V_k^+)^* V_k^+ = (V_k^-)^* V_k^-.$$

Use the Moore–Penrose pseudoinverse to set

$$W_k = V_k^+ G_k^\dagger (V_k^-)^*, \quad \tilde{\Gamma} = \sum_{k=0}^{m-1} W_k. \quad (45)$$

All these matrices are real; the adjoint notation is retained for convenience. The positive and negative spaces are two copies of $\mathbb{R}^\Omega$. Inputs with zero π_+ mass give zero rows, and inputs with zero π_- mass give zero columns, so no division by a possibly zero probability is involved.

Proposition 6.6 (Adversary conversion bounds). *The matrix in (45) satisfies*

$$\|\tilde{\Gamma}\| = m, \quad \tilde{\Gamma}\sqrt{\pi_-} = m\sqrt{\pi_+}, \quad \|\tilde{\Gamma} \circ \Delta_t\| \leq 2 \quad \text{for every entry coordinate } t.$$

For subsets $X, Y \subseteq \Omega$ with $\vartheta = \pi_+(X) - \pi_-(\Omega \setminus Y) > 0$,

$$\|\tilde{\Gamma}[X, Y]\| \geq m\vartheta/2.$$

Here brackets denote restriction to the indicated rows and columns.

Proof. We specialize the construction in [7, Theorem 15] and give the required norm estimates directly. All source numbering refers to the published TQC version. Its hypotheses are equality of assignment probabilities through size $2m$ on two finite-alphabet probability spaces. Here they hold because $\pi_\pm = 2\phi^\pm$ are probability measures and $2m \leq L_n$. The possibly different coordinate alphabets may be embedded in one finite label set; coordinate inequalities and assignment moments are unchanged.

The common-Gram identity above identifies our pseudoinverse formula with the source’s assignment isometry:

$$W_k V_k^- c = V_k^+ G_k^\dagger G_k c = V_k^+ c, \quad W_k|_{(\text{ran } V_k^-)^\perp} = 0.$$

Indeed, $\ker V_k^+ = \ker V_k^- = \ker G_k$, and $G_k^\dagger G_k$ is the orthogonal projection onto $(\ker G_k)^\perp$. Thus the displayed identity holds even when G_k is singular. It also shows that W_k is an isometry from $\text{ran } V_k^-$ onto $\text{ran } V_k^+$ and is zero on the orthogonal complement. Thus each W_k is a partial isometry, and $\|W_k\| \leq 1$. For the empty assignment $\emptyset$ one has $V_k^\pm e_\emptyset = \sqrt{\pi_\pm}$, so $W_k \sqrt{\pi_-} = \sqrt{\pi_+}$ for every k ; summing over k gives $\tilde{\Gamma} \sqrt{\pi_-} = m \sqrt{\pi_+}$. The triangle inequality gives $\|\tilde{\Gamma}\| \leq \sum_{k=0}^{m-1} \|W_k\| \leq m$, whereas the displayed unit-vector identity gives the reverse inequality. Hence $\|\tilde{\Gamma}\| = m$. These

isometries are compatible: on $\text{ran } V_k^-$, both W_k and W_m send each assignment vector to the corresponding positive vector. Thus W_k is the restriction of the common isometry W_m , extended by zero. Spans of assignments of size at most k equal those of size exactly k by summing extensions on unused coordinates (there are enough because $k \leq m < 3n^2$); thus both conventions give the same construction. Zero-probability rows and columns may be omitted or padded with zeros.

This recovers Theorem 15(a) of [7]. To verify the filtered bound, let P_k be the orthogonal projector onto $\text{ran } V_k^-$, so $W_k = W_m P_k$. For a coordinate t and $k \geq 1$, let $Q_{k,t}$ project onto the span of negative assignment vectors of size at most k whose assignments fix t , and set $Q_{0,t} = 0$. Summing an assignment vector over all possible values at t shows that

$$P_{k-1} \leq Q_{k,t} \leq P_k \quad (1 \leq k \leq m),$$

where the inequalities mean inclusion of ranges. Hence the projections $P_k - Q_{k,t}$, $0 \leq k \leq m-1$, have mutually orthogonal ranges. The range of $Q_{k,t}$ is an orthogonal direct sum over the fixed values of t . The isometry W_m takes each summand to positive assignment vectors supported on the same value. Thus $W_m Q_{k,t}$ vanishes whenever its row and column disagree at t . It follows that

$$B_t = W_m \sum_{k=0}^{m-1} (P_k - Q_{k,t})$$

has norm at most one and agrees with $\tilde{\Gamma}$ on all disagreement entries. For a rectangular matrix B , let $E_t(B)$ retain only entries whose row and column agree at coordinate t . This is a block-diagonal compression between mutually orthogonal coordinate-value subspaces, so $\|E_t(B)\| \leq \|B\|$. Consequently

$$\|\tilde{\Gamma} \circ \Delta_t\| = \|B_t - E_t(B_t)\| \leq 2 \|B_t\| \leq 2.$$

This is the norm-one representative of Theorem 15(b), equation (16), and the nonbinary filtering estimate of Proposition 4 in [7].

For completeness, set $a = \pi_+(X)$ and $b = \pi_-(\Omega \setminus Y)$, so $a - b = \vartheta > 0$. The restriction calculation is

$$\|\tilde{\Gamma}[X, Y]\| \geq m(\sqrt{a} - \sqrt{b}) \geq m\vartheta/2.$$

Indeed, let $u = \sqrt{\pi_-}|_Y$. Since $b < 1$, this vector is nonzero, and $\|u\| \leq 1$. Restricting $\tilde{\Gamma}\sqrt{\pi_-} = m\sqrt{\pi_+}$ to the rows in X gives

$$\begin{aligned} \|\tilde{\Gamma}[X, Y]u\| &\geq m \|\sqrt{\pi_+}|_X\| - \|\tilde{\Gamma}[X, \Omega \setminus Y](\sqrt{\pi_-}|_{\Omega \setminus Y})\| \\ &\geq m\sqrt{a} - m\sqrt{b}. \end{aligned}$$

Dividing by $\|u\| \leq 1$ proves the first inequality. The second is $\sqrt{a} - \sqrt{b} = (a-b)/(\sqrt{a} + \sqrt{b}) \geq (a-b)/2$. This is the restriction step of [7, proof of Theorem 17], applied here to the finite-alphabet matrix, and completes the specialization. $\square$

6.5 The singleton-residual strengthening

Proof of the adversary and promise assertions in Theorem 1.4. Set $X = f^{-1}(1)$, $Y = f^{-1}(0)$, and $Y_1 = \{AB - C = E_{ij} \text{ for some } i, j\} \subseteq Y$. The correction σ has total mass zero by the empty-assignment constraint and satisfies $\|\sigma\|_1 \leq \kappa = 1/16$; hence its positive and negative masses are each

at most $\kappa/2$. The seed has mass $1/2$ on X , is nonnegative there and on $\Omega \setminus Y_1$, and is nonpositive on Y_1 . Consequently

$$\begin{aligned}\lambda^+(X) &\geq \lambda(X) = 1/2 - \sigma(X) \geq 1/2 - \kappa/2, \\ \lambda^-(\Omega \setminus Y_1) &\leq \sigma^+(\Omega \setminus Y_1) \leq \kappa/2.\end{aligned}$$

Using $\pi_{\pm} = 2\lambda^{\pm}/\Lambda$ and $0 < \Lambda \leq 1 + \kappa$ gives

$$\pi_+(X) - \pi_-(\Omega \setminus Y_1) = \frac{2}{\Lambda}(\lambda^+(X) - \lambda^-(\Omega \setminus Y_1)) \geq \frac{1 - 2\kappa}{1 + \kappa} = \frac{14}{17}.$$

Proposition 6.6 therefore makes $\Gamma_1 = \frac{1}{2}\tilde{\Gamma}[X, Y_1]$ a valid adversary: the nonbinary filtered-norm bound two becomes one after scaling, and $\|\Gamma_1\| \geq 7m/34$. The correction need not be supported on the promise. Enlarging the NO set to all nonzero residuals supported on one entry permits zero-column padding, which preserves the operator norm and every filtered norm.

For general inputs define

$$\Gamma = \frac{1}{2}\tilde{\Gamma}[X, Y]. \quad (46)$$

Its filtered norms are at most one by the same proposition, and $\Gamma[X, Y_1] = \Gamma_1$. Since restriction cannot increase operator norm,

$$\text{Adv}^{\pm}(f) \geq \|\Gamma\| \geq \|\Gamma_1\| \geq \frac{7m}{34} = \frac{7}{34} \left\lfloor \frac{L_n}{2} \right\rfloor = \Omega(n^{5/3}). \quad (47)$$

Equations (38), (40), (42), (43), (44), (45), and (46), together with restriction to Y_1 , specify every matrix used. The adversary theorem supplies the corresponding query lower bounds. This proves the general-input and promise adversary assertions; the distributional part was proved in Corollary 6.3. $\square$

6.6 Adversary and promise bounds over unital rings

We now transfer the finite witnesses to arbitrary nonzero unital rings. No infinite-dimensional adversary theorem is needed.

Proof of Theorem 1.1: adversary bound. All coordinate maps in the two encodings of Section 5 are injective on their source alphabets. This is immediate in characteristic zero. In (31), the map $a \mapsto a1_R$ is injective on $\{0, \dots, p-1\}$ because $p \leq m$, and $a \mapsto ae$ is injective because e has order p . Writing Φ for either triple encoding, for every coordinate t we have

$$\mathbf{1}[\Phi(x)_t \neq \Phi(y)_t] = \mathbf{1}[x_t \neq y_t].$$

Relabel the rows and columns of a source adversary matrix by Φ . Its operator norm and every filtered norm remain identical. Thus the finite adversary witnesses themselves transfer, not merely the query lower bounds. Applying this preservation to the finite adversaries just constructed proves $\text{Adv}^{\pm}(\text{MPV}_R) = \Omega(n^{5/3})$: use the integer restriction in characteristic zero and the prime-field restriction in positive characteristic. The constants are absolute, and every witness has finite support, as required. $\square$

Singleton residuals. Here E_{ij} denotes the matrix with 1_R at (i, j) and zeros elsewhere. The characteristic-zero embedding preserves the promise residuals 0 and E_{ij} . In characteristic m , (32) instead takes a prime-field unit residual to eE_{ij} . Consequently the same lower bounds hold for the ring promise

$$AB - C \in \{0\} \cup \{eE_{ij} : i, j \in [n]\}.$$

When $m = p$ is prime, $e = 1_R$, so this is the unit-singleton promise, including for every positive-characteristic field. For composite m , this reduction alone does not establish the promise with residual $1_R E_{ij}$. For example, in $\mathbb{Z}/4\mathbb{Z}$ it uses $e = 2$ and the encoding $(A, B, C) \mapsto (\hat{A}, 2\hat{B}, 2\hat{C})$ of an $\mathbb{F}_2$ instance; $e^2 = 0$ is harmless because only B, C are scaled.

Other residual promises. Over the integers and fields, the unit-singleton theorem already implies hardness under the sole restriction that the residual have at most a specified positive number of nonzero entries, or rank at most one. More generally, any residual promise containing zero and every E_{ij} inherits the lower bound. This does not give the same conclusion when many errors are required, or when the factors themselves must be sparse or low rank. In composite characteristic, use eE_{ij} as above.

7 Rectangular verification and consequences

We extend the transfer method to rectangular products and record its localization consequence.

7.1 Rectangular products with a long inner dimension

Retaining the inner dimension while inspecting a smaller output block yields more than padding a square hard instance.

Corollary 7.1 (Square-output rectangular verification). *Let $1 \leq s \leq k$ and let R be a fixed nonzero unital associative ring. For entry-oracle matrices*

$$A \in R^{s \times k}, \quad B \in R^{k \times s}, \quad C \in R^{s \times s},$$

deciding whether $AB = C$ has bounded-error worst-case query complexity

$$Q_{1/3} = \Theta(ks^{2/3}).$$

The lower bound has finite support and absolute constants, independent of R . Over finite fields it is uniform in the field size; over the integers it holds on

$$\mathcal{X}_{s,k,H}^{\text{rect}} = I_H^{s \times k} \times I_H^{k \times s} \times J_{k,H}^{s \times s}, \quad H \geq H_k,$$

uniformly in H . Choosing $H = H_k = 2^{100k}$ gives $O(k)$ bits per entry. The same finite-domain lower bounds hold for bounded approximate coordinate degree.

Proof. We first prove the lower bound over a finite field and over the integer restriction in the statement. Let p be a globally $[0, 1]$ -valued function of coordinate degree d on the rectangular domain that is within $1/3$ of the rectangular predicate on every input of that domain. Apply the majority amplifier Maj_{s_0} from the proof of Proposition 2.7. The resulting function p_* has error at most 0.01, remains globally in $[0, 1]$, and has degree at most $L = s_0 d$. If $L > k^{5/3}$, then already $d > k^{5/3}/s_0 \geq ks^{2/3}/s_0$, so assume $L \leq k^{5/3}$.

Let π project a square k -dimensional triple onto the first s rows of its first factor, the first s columns of its second factor, and the upper-left s -by- s block of its proposed product. Define $P = p_* \circ \pi$. In the integer case, π maps $\mathcal{X}_{k,H}$ onto $\mathcal{X}_{s,k,H}^{\text{rect}}$; in the field case it maps the full square product domain onto the full rectangular product domain. Consequently $P \in V_L$ and $0 \leq P \leq 1$ on the entire square domain, not just on inputs used by the residual distribution.

For $D \in \{0,1\}^{s \times s}$ let $\iota(D)$ be the k -by- k matrix with upper-left block D and zeros elsewhere. Independence of the sampled entries gives the exact identity

$$(\mathcal{R}_k P)(\iota(D)) = \mathbb{E}_{A,Bp_*}(A, B, AB - D), \quad (48)$$

where $\mathcal{R}_k$ denotes the square residual average at dimension k , and on the right A, B have the rectangular shapes in the statement. Their entries are independent and uniform in the field, or in $[-H, H] \cap \mathbb{Z}$, respectively. In the integer case all entries of $AB - D$ lie in $J_{k,H}$, so every right-hand input belongs to the stated product restriction.

Apply Theorem 3.3 or Theorem 4.5 at dimension k , and then restrict its output along ι . For the field theorem also restrict each remaining residual coordinate to $\{0,1\}$. Each term depending on at most K rows and K columns retains this property under either restriction, so the corresponding linear span is preserved. Thus we obtain a real function $h_0 \in \mathcal{W}_K$ on $\{0,1\}^{s \times s}$, where

$$K = \lfloor 16L/k \rfloor, \quad |h_0(D) - \mathbb{E}_{A,Bp_*}(A, B, AB - D)| \leq \eta_k, \quad \eta_k = e^{-c_* k/2}.$$

For all sufficiently large k the estimate is uniform in $s \leq k$, in the field size, and in $H \geq H_k$. By (48), the averaged function is at least 0.99 at zero and at most 0.01 at every nonzero Boolean residual. For $\eta_k \leq 1/200$, normalization as in Corollary 2.4 gives

$$h = \frac{h_0 + \eta_k}{1 + 2\eta_k} \in \mathcal{W}_K, \quad 0 \leq h \leq 1, \quad h(0) \geq 0.98, \quad h(D) \leq 0.02 \quad (D \neq 0).$$

Lemma 2.6, with residual dimension s , gives $K \geq c_1 s^{2/3}$ for every $s \geq 1$. Hence

$$16s_0 d/k \geq K \geq c_1 s^{2/3}, \quad d = \Omega(ks^{2/3}).$$

The bounded range of dimensions excluded by the transfer threshold is absorbed by reducing the absolute constant: the predicate is nonconstant and therefore has degree and bounded-error query complexity at least one. The proof of Lemma 2.2 applies to the $2sk + s^2$ addressed entries, so the coordinate-degree bound yields the query lower bound. For an arbitrary fixed R , use the characteristic-zero embedding or the asymmetric positive-characteristic encoding (31). Their residual identity (32) is valid for rectangular products as well, and each encoded query uses a constant number of source queries. Composition with either coordinatewise encoding cannot increase coordinate degree. Thus both lower bounds transfer to the corresponding finite restrictions over R .

For the upper bound, apply the stored-submatrix bound (33) with $t = s \leq k$, obtaining $O(ks^{2/3})$ queries, including the direct-reading case $s < 8$. The exact cached check applies to every ring in the statement. $\square$

This corollary settles the regime of square outputs with inner dimension at least the output side. The upper bound extends: the walk of Section 5 costs $O(kr + kt/\sqrt{r})$ queries whenever $1 \leq r \leq \min\{k, t/2\}$, with bounded t handled by direct reading. Thus $O(ks^{2/3})$ queries suffice for all $s \leq k^{3/2}$, matching the hypothesis of [8, Section 6.1]; only the lower-bound argument uses $s \leq k$. The corollary does not assert a sharp bound for arbitrary m -by- k times k -by- p shapes.

7.2 Locating an erroneous entry

Corollary 7.2 (Locating an error). *Over every fixed nonzero unital associative ring, the task of returning `equal` when $AB = C$, and otherwise returning an index (i, j) with $(AB)_{ij} \neq C_{ij}$, has bounded-error worst-case query complexity $\Theta(n^{5/3})$.*

Proof. The lower bound follows from decision. For the upper bound, use the rectangular-verification and binary-search idea of Buhrman and Špalek [8]. For a t -by- t output block with inner dimension n and $t \leq 2n$, the stored-submatrix bound (33), with $k = n$, gives $O(nt^{2/3})$ queries, including the direct-reading case $t < 8$.

Pad the output to a square of power-of-two side less than $2n$, using constant zero rows and columns. Test the whole output and return `equal` if that test declares equality. Otherwise, recursively test the four quadrants and retain one declared erroneous. At depth ℓ , make each test’s error at most $2^{-\ell-8}$ by $O(\ell + 1)$ repetitions, using fresh workspaces and randomness for each test. Conditional on any preceding history, the next chosen block therefore has the same worst-case error guarantee. There are at most four tests per depth, so the total error probability is at most $4 \sum_{\ell \geq 0} 2^{-\ell-8} < 1/3$, also for the adaptively chosen blocks. If no quadrant is declared erroneous, return `equal`. At a leaf, read its dot product exactly: return its index if it is wrong, and `equal` otherwise. The query cost, including this final $O(n)$ check, is

$$O\left(n^{5/3} \sum_{\ell \geq 0} (\ell + 1) 2^{-2\ell/3} + n\right) = O(n^{5/3}).$$

No false witness is output; all failures to locate a genuine error are contained in the union of the verification-error events. $\square$

Scope and limitations. All bounds in this paper concern exact whole-entry access. We do not address approximate verification, bit-query complexity, noisy or finite-precision access. The integer construction uses $O(n)$ -bit entries. Whether the same lower bound holds on an integer restriction with $O(\log n)$ bits per entry, or with all entries from some fixed finite subset of $\mathbb{Z}$ independent of n , remains open. Rectangular shapes outside Corollary 7.1, dependence on the number of wrong entries, listing or correcting all errors, and nonunital rings are not classified here. Ordinary products over $\mathbb{Z}$ or $\mathbb{F}_2$ and Boolean OR–AND products are different problems: for the Boolean semiring the cited works give an $O(n^{3/2})$ upper bound and a $\tilde{\Omega}(n^{19/18})$ lower bound, where the tilde suppresses polylogarithmic factors [8, 10, 15]. None of our reductions above covers Boolean or min-plus semirings, or stronger row, matrix-vector, or block-encoding access.

Use of AI: The author started with this prompt to solve the *finite fields case* of matrix-product verification: “Use polynomial method. Average degree- L acceptance polynomial over random matrix factors and restrict its residual Fourier spectrum to $O(L/n)$ active rows and columns. Use element-distinctness hardness to force it to $\Omega(n^{2/3})$ and derive $L = \Omega(n^{5/3})$ ”. ChatGPT generated an outline. The author fixed some missing gaps and iteratively verified and expanded the proofs with ChatGPT and prompted it to work out the details of the adversary bounds and also apply the techniques to other input types and variants of the *matrix-product verification problem*. The author is solely responsible for the correctness of the proofs.

References

- [1] Scott Aaronson and Yaoyun Shi. Quantum lower bounds for the collision and the element distinctness problems. *Journal of the ACM*, 51(4):595–605, 2004. [doi:10.1145/1008731.1008735](https://doi.org/10.1145/1008731.1008735).
- [2] Iskander Aliev, Martin Henk, and Eva Linke. Integer points in knapsack polytopes and s -covering radius. *Electronic Journal of Combinatorics*, 20(2), Paper P42, 2013. [doi:10.37236/2887](https://doi.org/10.37236/2887).
- [3] Andris Ambainis. Polynomial degree and lower bounds in quantum complexity: Collision and element distinctness with small range. *Theory of Computing*, volume 1, pages 37–46, 2005. [doi:10.4086/toc.2005.v001a003](https://doi.org/10.4086/toc.2005.v001a003).
- [4] Andris Ambainis. Quantum walk algorithm for element distinctness. *SIAM Journal on Computing*, 37(1):210–239, 2007. [arXiv:quant-ph/0311001](https://arxiv.org/abs/quant-ph/0311001).
- [5] Keith Ball. Cube slicing in $\mathbb{R}^n$. *Proceedings of the American Mathematical Society*, 97(3):465–473, 1986. [doi:10.1090/S0002-9939-1986-0840631-0](https://doi.org/10.1090/S0002-9939-1986-0840631-0).
- [6] Robert Beals, Harry Buhrman, Richard Cleve, Michele Mosca, and Ronald de Wolf. Quantum lower bounds by polynomials. *Journal of the ACM*, 48(4):778–797, 2001. [arXiv:quant-ph/9802049](https://arxiv.org/abs/quant-ph/9802049).
- [7] Aleksandrs Belovs. A direct reduction from the polynomial to the adversary method. In *19th Conference on the Theory of Quantum Computation, Communication and Cryptography (TQC 2024)*, LIPIcs 310, Article 11, 2024. [doi:10.4230/LIPIcs.TQC.2024.11](https://doi.org/10.4230/LIPIcs.TQC.2024.11). See also [arXiv:2301.10317](https://arxiv.org/abs/2301.10317).
- [8] Harry Buhrman and Robert Špalek. Quantum verification of matrix products. In *Proceedings of the Seventeenth Annual ACM–SIAM Symposium on Discrete Algorithms (SODA)*, pages 880–889, 2006. [arXiv:quant-ph/0409035](https://arxiv.org/abs/quant-ph/0409035).
- [9] Mark Bun and Justin Thaler. Dual polynomials for collision and element distinctness. *Theory of Computing*, 12(16):1–34, 2016. [doi:10.4086/toc.2016.v012a016](https://doi.org/10.4086/toc.2016.v012a016).
- [10] Andrew M. Childs, Shelby Kimmel, and Robin Kothari. The quantum query complexity of read-many formulas. In *Algorithms—ESA 2012*, LNCS 7501, pages 337–348. Springer, 2012. [doi:10.1007/978-3-642-33090-2_30](https://doi.org/10.1007/978-3-642-33090-2_30).
- [11] Rūsiņš Freivalds. Fast probabilistic algorithms. In *Mathematical Foundations of Computer Science (MFCS)*, LNCS 74, pages 57–69. Springer, 1979. [doi:10.1007/3-540-09526-8_5](https://doi.org/10.1007/3-540-09526-8_5).
- [12] Peter Høyer, Troy Lee, and Robert Špalek. Negative weights make adversaries stronger. In *Proceedings of the Thirty-Ninth Annual ACM Symposium on Theory of Computing (STOC)*, pages 526–535, 2007. [doi:10.1145/1250790.1250867](https://doi.org/10.1145/1250790.1250867).
- [13] Wassily Hoeffding. Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association*, 58(301):13–30, 1963. [doi:10.1080/01621459.1963.10500830](https://doi.org/10.1080/01621459.1963.10500830).
- [14] Yuki Kelly Itakura. Quantum algorithm for commutativity testing of a matrix set. MMath essay, University of Waterloo, 2005. [arXiv:quant-ph/0509206](https://arxiv.org/abs/quant-ph/0509206).

- [15] Robin Kothari and Ashwin Nayak. Quantum algorithms for matrix multiplication and product verification. In Ming-Yang Kao, editor, *Encyclopedia of Algorithms*, pages 1673–1677. Springer, 2016. doi:[10.1007/978-1-4939-2864-4_303](https://doi.org/10.1007/978-1-4939-2864-4_303).
- [16] Troy Lee, Rajat Mittal, Ben W. Reichardt, Robert Špalek, and Mario Szegedy. Quantum query complexity of state conversion. In *Proceedings of the Fifty-Second IEEE Symposium on Foundations of Computer Science (FOCS)*, pages 344–353, 2011. doi:[10.1109/FOCS.2011.75](https://doi.org/10.1109/FOCS.2011.75).
- [17] Frédéric Magniez, Ashwin Nayak, Jérémie Roland, and Miklós Santha. Search via quantum walk. *SIAM Journal on Computing*, 40(1):142–164, 2011. doi:[10.1137/090745854](https://doi.org/10.1137/090745854).
- [18] James Melbourne and Cyril Roberto. Quantitative form of Ball’s cube slicing in $\mathbb{R}^n$ and equality cases in the min-entropy power inequality. *Proceedings of the American Mathematical Society*, 150(8):3595–3611, 2022. doi:[10.1090/proc/15944](https://doi.org/10.1090/proc/15944). See also [arXiv:2109.03946](https://arxiv.org/abs/2109.03946).
- [19] Giacomo Micheli and Violetta Weger. On rectangular unimodular matrices over the algebraic integers. *SIAM Journal on Discrete Mathematics*, 33(1):425–437, 2019. doi:[10.1137/18M1177093](https://doi.org/10.1137/18M1177093). See also [arXiv:1803.08785](https://arxiv.org/abs/1803.08785).
- [20] Jacob T. Schwartz. Fast probabilistic algorithms for verification of polynomial identities. *Journal of the ACM*, 27(4):701–717, 1980. doi:[10.1145/322217.322225](https://doi.org/10.1145/322217.322225).
- [21] Robert Špalek and Mario Szegedy. All quantum adversary methods are equivalent. *Theory of Computing*, 2(1):1–18, 2006. doi:[10.4086/toc.2006.v002a001](https://doi.org/10.4086/toc.2006.v002a001).
- [22] Mario Szegedy. Quantum speed-up of Markov chain based algorithms. In *Proceedings of the Forty-Fifth IEEE Symposium on Foundations of Computer Science (FOCS)*, pages 32–41, 2004. doi:[10.1109/FOCS.2004.53](https://doi.org/10.1109/FOCS.2004.53).